

Artificial Intelligence in K-12 Education: An Umbrella Review

Ruiping Huang ^a, Yue Yin ^a, Na Zhou ^a, Fei Lang ^{a,b}

^a College of Education, University of Illinois Chicago, USA

^b School of Foreign Languages, Harbin University of Science and Technology, China

Corresponding author and address

Ruiping Huang rhuang26@uic.edu

College of Education, University of Illinois Chicago

1040 W Harrison St, Chicago, IL 60607

Coauthors:

Yue Yin yueyin@uic.edu

College of Educational, University of Illinois Chicago

1040 W Harrison St, Chicago, IL 60607

Na Zhou nzhou25@uic.edu

College of Education, University of Illinois Chicago

1040 W Harrison St, Chicago, IL 60607

Fei Lang langfei@gmail.com

College of Education, University of Illinois Chicago, USA

School of Foreign Languages, Harbin University of Science and Technology, China

Statement on open data and ethics

The data supporting the findings of this study are available from the corresponding author upon reasonable request. As this study is an umbrella review, it does not report or involve the use of any animal or human data, and therefore did not require ethics approval or informed consent.

CRedit authorship contribution statement

Ruiping Huang: Conceptualization, Data curation, Formal analysis, Methodology, Coding for data traction and quality assessment, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing, Managing references.

Yue Yin: Conceptualization, Writing – original draft, Writing – review & editing, Coding for data extraction and quality assessment, Visualization, Project administration, Supervision, Funding acquisition.

Na Zhou: Writing – review & editing, Writing – original draft, Coding for data extraction and quality assessment, Managing references.

Fei Lang: Writing – review & editing, Coding for data extraction and quality assessment, Visualization.

Acknowledgments

This paper is based upon work supported by the National Science Foundation (NSF) under Grant Number DRL-2101166. However, the opinions, findings, and conclusions are those of the authors and do not necessarily reflect the views of the NSF. The authors have received various contributions from Jeremy Riel, Ying Chen, and Sania Zaidi during the preparation of this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to proofread grammars. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of interest statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Artificial Intelligence in K-12 Education: An Umbrella Review

Ruiping Huang ^a, Yue Yin ^a, Na Zhou ^a, Fei Lang ^{a,b}

^a College of Education, University of Illinois Chicago, USA

^b School of Foreign Languages, Harbin University of Science and Technology, China

Highlights

- This umbrella review summarizes 102 systematic reviews on Artificial Intelligence in Education (AIEd) in K-12.
- We created a conceptual AIEd review framework to map the territory of AIEd and synthesize the major findings in the systematic reviews.
- The existing systematic reviews mostly focus on AI applications but lack attention to AI education and literacy, and AIEd theories.
- Many review studies need to improve their data management, extracting process, analysis method, and the report of limitations.
- We developed a quality assessment rubric for umbrella review, which can be used to evaluate systematic reviews across various fields.

Abstract

This umbrella review examines the current state of artificial intelligence in education (AIEd) in K-12 by synthesizing 102 systematic reviews. In this umbrella review, we describe the characteristics of the reviews on AIEd, build a framework to summarize the review focuses, report the major findings in the reviews, and examine the quality of these reviews. The reviews have made a substantial contribution by synthesizing the advancement in AI education and development. Meanwhile, future work may focus more on the following directions: a) reviews on AI education and literacy; b) reviews on the empirical utilization of AI tools in education, c) more reviews on the theories used in AIEd, and (d) improving data management, extraction process, analysis method, and the report of limitations for systematic reviews. This umbrella review maps the territory of AIEd, synthesizes the major knowledge obtained through systematic reviews on AIEd, and identifies future directions. The framework we created can be used to guide future reviews in AIEd. The quality assessment rubric we refined can be used for future umbrella reviews in general.

Keywords

Artificial intelligence; K-12 Education; Umbrella review; Systematic review

Acronyms

- AI: Artificial Intelligence
- AIEd: Artificial Intelligence in Education
- AR: Augmented Reality
- ICC: Intra-class Correlation Coefficient
- PAs: Pedagogical Agents
- PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses
- RQ: Research Questions
- VR: Virtual Reality

Databases and digital libraries

- ACM Digital Library: Association for Computing Machinery's digital library
- APA PsycInfo: American Psychological Association's Psychological Information Database
- Education Research Complete: A comprehensive database of education-related research (EBSCOhost)
- ERIC: Education Resources Information Center
- IEEE Xplore: Institute of Electrical and Electronics Engineers' digital library
- ScienceDirect: Elsevier's database platform for scientific and technical content
- Web of Science: Clarivate's Web of Science Core Collection

Artificial intelligence (AI) is rapidly developing and playing an increasingly important role in our daily lives (Tahiru, 2021). AI technologies have been applied in education, serving as intelligent tutors, tutees, tools/partners, and policymaking advisors in educational settings to facilitate teaching, learning, and decision making (Hwang et al., 2020). The number of studies and reviews on AI in education (AIEd) has dramatically increased (Chen et al., 2020a) in the past decades. However, no sufficient umbrella reviews systematically consolidate the existing systematic reviews and examine their qualities. We conducted an umbrella review, also known as a review of review (Grant & Booth, 2009), to synthesize the systematic reviews on AIEd in K-12 and examine their quality to guide future endeavors.

1. Literature review

1.1. The Development of AIEd

AI has been at the forefront of technological advancements for decades, significantly enhancing human capabilities (Kelly et al., 2023). Scholarly literature offers many definitions of AI. McCarthy (2006) defined AI as "the science and engineering of creating intelligent devices, particularly intelligent computer programs." Over time, AI has been broadly recognized as a set of computational systems and tools (Southworth et al., 2023). Kelly et al. (2023) offer a more specific definition, describing AI as an artificial entity capable of meeting or exceeding task requirements while accounting for cultural and demographic contexts.

In K-12 education, AI has the potential to facilitate assessment, monitor and track students' progress, predict students' and teachers' success, personalize learning, diagnose learning gaps, and deliver and generate knowledge (Xu & Ouyang, 2022a; Chen et al., 2020b; Zafari et al., 2022; Martin et al., 2024). With the surge in interest surrounding AIEd, articles on this topic have skyrocketed in recent years. According to the Web of Science database, the number of AIEd articles or review articles increased from 414 in 2020 to 3,838 in 2024.

1.2. AIEd Reviews

With the dramatic increase of publications in AIEd, a substantial number of reviews have been published as well. The Web of Science database shows 1155 reviews were published in the five-year period from 2020 to 2024. Researchers have reviewed AIEd studies from various perspectives, such as AI education, applications, and AI technology development, challenges, and ethics. Below are some examples.

Given the importance of AI education, many reviews focus on AI related education. For example, Gennari et al. (2023) reviewed tools designed to teach AI to K-12 learners in an engaging and playful manner. Their analysis highlighted that tools are available for students across all grade levels, with digital tools being the most extensively studied and utilized. Su et al.

(2024) reviewed the AI effects based on the AI literacy framework developed by Ng et al. (2021), which were knowing and understanding AI, using and applying AI, evaluating and creating AI, and AI ethics. They found that participating in AI curricula enabled students to learn the basic functions of AI, apply AI knowledge, evaluate and create AI applications, and understand AI-related ethical issues.

A large number of reviews summarizes AI applications (e.g., Dai & Ke, 2022; Zafari et al., 2022). Some summarize the overall AI applications, e.g., Suryanti et al. (2024) reviewed and divided AI Application research into seven categories: learning personalization, virtual tutor, learning surveillance, learning assistant, learning facilities, automatic assessment, and voice assistant. Other reviews focus on a specific AI application, such as, automated essay scoring (Ramesh, 2022), science assessment (Zhai et al., 2020), and second-language learning (Almelhes, 2023).

Reviews on AIEd also encompass AI development for education from a technical perspective. Researchers have reviewed the design and evaluation of AI algorithms (e.g., Mousavinasab et al., 2021; Tan et al., 2022; Orji & Vassileva, 2022), the analysis and application of programming languages (e.g., Bhutoria, 2022), and the operating platforms (e.g., Kuhail et al., 2023) that support AI applications in K-12 education context.

Reviews also synthesize various challenges in AIEd. For example, Saputra et al. (2023) identified key challenges in education, including pedagogical limitations, misaligned frameworks, insufficient AI literacy, ethical concerns, impacts on character development, high costs, inadequate teacher training, and slow curriculum adaptation. Su et al. (2023) highlighted several challenges to AI literacy: a lack of teachers' AI knowledge, skills, and confidence; limited curriculum design; and the absence of clear teaching guidelines. From a technical perspective, Cuellar-Rojas et al. (2022) found that it is challenging to ensure data privacy and security, mitigate algorithmic biases, address limitations in datasets, and overcome technological and infrastructural barriers such as high costs and resource constraints.

Some studies examined the ethical concerns on AIEd. For instance, Vargas-Murillo (2023) raised the concerns due to potential misuse of AIEd. For example, AI infiltrating academic papers endanger the integrity and credibility of researchers, AI undermines creative thinking, and students may submit AI generated writing as their own work, leading to academic dishonesty.

1.3. Umbrella Review

While these reviews have provided valuable insights and significantly contributed to the understanding of AIEd, it remains unclear what has been systematically reviewed, what needs further review, and the quality of existing reviews. When extensive primary research and a

growing number of reviews have been conducted on a topic, an umbrella review is a logical next step to advance knowledge and improve outcomes (Conn & Coon Sells, 2014).

An umbrella review synthesizes and evaluates findings from multiple systematic reviews and meta-analyses on a broad topic (Aromataris et al., 2015; Grant & Booth, 2009). It can locate, appraise, and synthesize systematic review level evidence (Becker & Oxman, 2008; Grant & Booth, 2009), provide a comprehensive examination and high-level synthesis of existing knowledge, highlight gaps (Gosling et al., 2023), assess the quality of existing reviews, and provide direction for future research (Conn & Coon Sells, 2014).

To meet the needs, we conducted an umbrella review to synthesize the systematic reviews on AIED in K-12 with the following focuses: First, we aimed to analyze the general characteristics of existing reviews, including publication year, journals, funding support, study count, inclusion of empirical studies, first author's country/region, educational settings, and education delivery modes. These analyses provided insights into the regional and temporal trends of AIED research, highlighted influential journals and funding patterns, and identified common research contexts and contributing countries that may reflect broader educational priorities and AI integration efforts.

Second, we aimed to examine the focus of these reviews, such as AI literacy, educational applications, ethical considerations, and technology development within K-12 environments. By classifying and analyzing the focuses of the systematic reviews, we developed a framework to capture the reviews on AIED, identified which aspects have been extensively studied and which areas remain underexplored, and thematically synthesized each focus by identifying key findings in the reviews.

Third, we aimed to assess the quality of these reviews. Quality assessment plays a critical role in evaluating the methodological rigor and risk of bias in included studies (Munn et al., 2023), ensuring study validity and reliability using standardized tools and checklists (Negarandeh & Beykmirza, 2020). By systematically assessing the quality of systematic reviews on AIED, we aimed to identify the gaps and opportunities for methodology improvement.

Accordingly, we addressed the following research questions (RQ) in our umbrella review:

- (a) RQ1. What are the characteristics of the systematic reviews on AIED in K-12?
- (b) RQ2. What are the review focuses of the systematic reviews on AIED in K-12?
- (c) RQ3. What are the quality assessment results of the systematic review studies on AIED in K-12?

By answering these questions, we aim to provide a comprehensive overview of AI in K-12 education, identify key trends, and highlight the gaps and limitations. This approach offers a holistic perspective on the current state of AI in K-12 education, informs scholars entering the AI field, and guides future endeavors in this rapidly evolving area.

2. Method

To achieve the goals, we conducted an umbrella review and managed the review process using Covidence, a systematic review data management software (<https://www.covidence.org/>). During the review process, we followed the widely adopted Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) checklist (Page et al., 2021). This section presents the search strategy, inclusion and exclusion criteria, screening process, data extraction and quality assessment, and data analysis methods. Fig. 1 illustrates the searching and screening processes exported from “Covidence”.

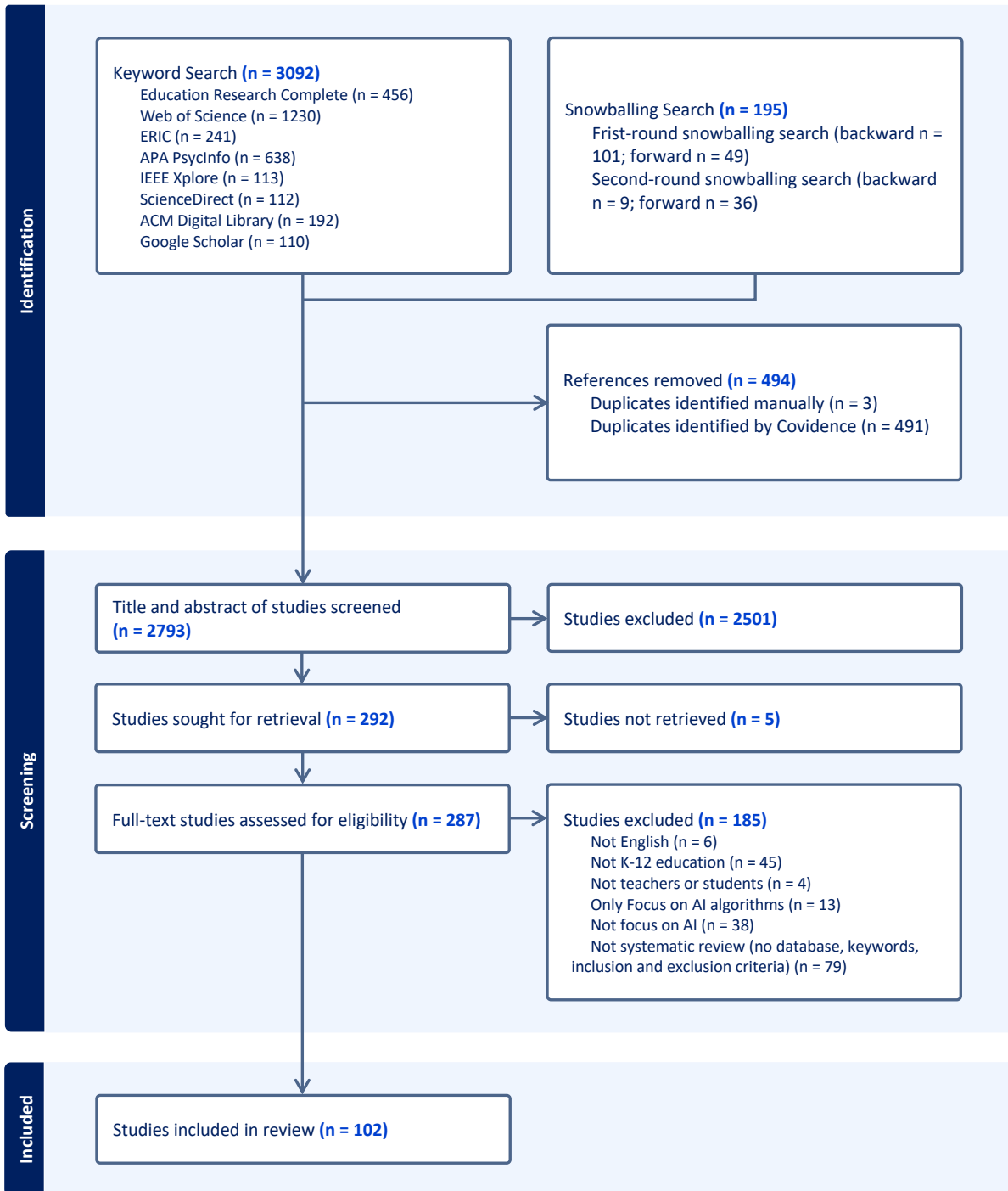


Fig. 1. PRISMA diagram.

2.1. Search Strategy and Study Election

Keyword Search. In September 2023, the first author searched seven databases: Education Research Complete, Web of Science, ERIC, APA PsycINFO, IEEE Xplore, ScienceDirect, and the ACM Digital Library. These databases were chosen for their relevance to education and AI to ensure comprehensive coverage in our systematic review. We used keywords related to AI and education, (artificial intelligence OR ai OR a.i. OR machine learning OR deep learning OR artificial thinking OR intelligent machines) AND (learning OR teaching OR students OR teachers OR educa*) AND (review). The initial search yielded 2982 articles. Given the rapid developments in AI and the emergence of new evidence, we updated our search in September 2024 by adding 110 review articles published between 2023 and 2024 from Google Scholar. Totally, we found 3092 articles using the keyword search approach.

Snowballing Search. We further conducted a snowballing search, because combining snowballing with keyword search is more effective than keyword search alone for identifying relevant primary studies (Wohlin et al., 2022). Given the large number of papers found through the keyword search ($n = 3,092$), we followed Wohlin's (2014) recommendation to identify a set of relevant and highly cited papers as the starting seed set. After screening the keyword search results, we selected 101 papers as the seed set. Using these papers, we automated the forward and backward snowballing (Wohlin, 2014) with Research Rabbit (<https://www.researchrabbit.ai/>), a free AI tool for literature discovery. After two rounds of snowballing searches, we identified no more additional qualified papers. Therefore, the iterative process ended, resulting in the selection of 195 papers through the snowballing approach.

2.2. Inclusion and Exclusion Criteria

We generated inclusion and exclusion criteria from the iteration of team brainstorming, discussion, and pilot testing (Table 1). Two reviewers screened each paper independently in Covidence and included reviews that meet the following criteria: (a) focusing on teachers and students, (b) systematic reviews, meta-analyses, or scoping reviews; (c) focusing on K-12 education; (d) focusing on AI applications, impacts, interpretations, education, or technologies; (e) peer-reviewed journal articles, IEEE conference papers, and ACM conference papers; (f) English articles.

Table 1

Inclusion and exclusion criteria.

Criteria	Inclusion	Exclusion
Target Population	Teachers (including pre-service and in-service teachers	Not teachers and students (e.g. school managers, and parents)

	who work for K-12 students) and students	
Study Design	Systematic review / Meta-analysis / Scoping review (reviews that show at least the three elements: searching database, search terms, and inclusion and exclusion criteria)	Not systematic review (reviews that don't show the three elements: searching database, search terms, and inclusion and exclusion criteria)
Educational Setting	Reviews focus on K -12 learning and teaching (including early childhood education, unspecific or general educational field)	Not focus on K-12 educational field (e.g. family education, educational management such as AI-based Proctoring Systems, MOOC, Distance education, higher education, large testing organizations such as TOEFL, AI applied in general life)
Review Focus	Reviews focus on AI applications, or impact, or interpretation, or education (e.g. teaching AI), or technologies.	1. Not focused on AI or only partially focused on AI: (1) Reviews that do not report on any aspect of AI (applications, impact, interpretation, education, or technologies). (2) Reviews that focus on topics like learning analytics, data mining, engagement detection systems, augmented reality (AR), virtual reality (VR), or the metaverse, which may incorporate AI partially or not at all, rather than having AI as the central focus. 2. Purely technology-based reviews Reviews that solely focus on AI algorithms without discussing their educational applications or implications.
Source of Publication	Peer-reviewed journals, IEEE conference papers, ACM conference papers	White papers, book chapters, technical reports, dissertations, empirical studies
Language	English	Not English

2.3. Screening Process

After we selected papers and removed the duplicates ($n = 493$), we screened the remaining papers in two steps. First, the first and second authors independently screened the titles and abstracts of 2793 articles. The percentage agreement is 91.14%, which was interpreted as excellent inter-rater reliability (Barth et al., 2017); Cohen's kappa is 0.62, which was interpreted as substantial inter-rater reliability (Landis & Koch, 1977; Gisev et al., 2013). We

resolved the discrepancies through group discussions with at least three team members, resulting in 287 articles for full text screening.

Second, the same two authors screened 287 papers independently based on the full texts. The percentage agreement is 83.43%, which was interpreted as good inter-rater reliability (Barth et al., 2017); Cohen's kappa is 0.66, which was interpreted as substantial inter-rater reliability (Landis & Koch, 1977; Gisev et al., 2013). We resolved the conflict decisions through group discussion, resulting in a final pool of 102 studies (Appendix A).

2.4. Data Extraction and Quality Assessment

The first author drafted the initial data extraction rubrics (Table 2) based on the research questions. We drafted our quality assessment rubrics using the PRISMA 2020 item checklist (Page et al., 2021) and its criteria established by Phuong et al. (2020). The research team then collaboratively refined these rubrics, pilot-tested them with 50 papers through discussion and iterative revisions, and finalized them when no more revisions were needed.

Using the finalized rubrics, we coded the review articles to answer the three research questions: (a) RQ1, Characteristics (Table 2): publication year, journal, funding support, study count, empirical studies inclusion, country/region, educational types, education delivery modes. (b) RQ2, Review Focuses (Table 2): AI education and literacy, AI applications, AI challenges, AI techniques development, and AIEd theories. (c) RQ3, Quality Assessment: Rationale, objectives, databases, strategy, criteria, management, screening, extracting, coding, methods, presentation, results, and limitations. Most dimensions include three level ratings: not stated, somewhat stated, and clearly stated.

Using the finalized rubrics, the first and third authors independently coded 50 (49.02%) randomly selected studies to establish inter-rater reliability and resolved conflicts through group discussion. For extraction rubrics (Table 2), the percentage agreement was 91.03%, and Cohen's kappa was 0.82, indicating excellent reliability (Barth et al., 2017; Landis & Koch, 1977). For quality assessment (13 ordinal items), we examined percentage agreement (82.46%) and intra-class correlation coefficient [ICC(C, 1)] (McGraw & Wong, 1996) at 0.62 (95% CI [0.57, 0.66]), indicating good to moderate reliability (Barth et al., 2017; Koo & Li, 2016). With satisfactory reliability, the first author independently coded the remaining papers, resolving ambiguities through group discussion to ensure consistency and accuracy.

Table 2

Extraction rubric.

Checklist Item	Choices
1. Publication Year	Text
2. Journal	Text
3. Funding Support	Text
4. Study Count (the number of studies finally included in the review)	Text
5. Empirical Studies Inclusion (only Empirical Studies Included?) [Single Choice]	• Yes (Only included empirical studies) • No (Included empirical and non-empirical studies) • Unspecified (didn't mention)
6. Country/Region (the country/region of the first author) [Multiple Choice]	Text
7. Educational Settings [Multiple Choice]	• Formal Education • Informal Education (e.g. community-based clubs, and extracurricular programs) • Unspecified (didn't mention) • Other
8. Education Delivery Modes [Multiple Choice]	• Online Learning (E-Learning) • Traditional Classroom-Based Learning • Hybrid or Blended Learning • Unspecified • Other
9. Review Focus (obtained from the research purpose, questions, results, or conclusions) [Multiple Choice]	• AI Education and Literacy: AI Education, AI Literacy, AI Ethics • AI Applications: AI Functions, AI Utilization Process • AI Challenges • AI Technology Development: AI Technologies, AI Programming Languages, AI Operating Platform • AIED Theories

2.5. Data Analysis

We systematically examined the selected reviews quantitatively and qualitatively. Quantitatively, we calculated frequencies and proportions for categorical variables, such as publication year, funding support, and educational modes, and presented the major results through tables and figures. Qualitatively, we synthesized themes and patterns by categorizing the review focuses into a multi-level framework to map the AIEd K-12 territory. By combining numerical rigor with contextual depth, we identified the trends and gaps, and recommended how to advance future research on AIEd K-12.

3. Results

3.1. RQ1: Characteristics of the Studie

We described the characteristics of the reviews from multiple dimensions: (a) The year of publication--The number of reviews constantly increased from 2020 to 2023, peaking at 43 in 2023 (Fig. 2). (b) Journal--The reviews appeared in 57 journals, with 12 journals publishing more than one reviews (Fig. 3). (c) Funding support--Forty-three (42.16%) studies received funding support. (d) The number of studies included--The number of studies included had a mean of 69.55 (SD = 85.67) (Fig. 4). (e) Inclusion of empirical studies--Most reviews (n=59, 57.84%) included only empirical studies, while 13 (12.75%) included both empirical and non-empirical studies, and 30 (29.41%) did not specify. (f) Country/region of the first author--The top three common country/region for the first author (Fig. 5) were China (n=16, 15.69%), Hong Kong (n=11, 10.78%), and USA (n=10, 9.80%). (g) Educational setting--Seven papers (6.86%) focused on formal education, ten (9.80%) on both formal and informal education, and 85 (83.33%) did not specify it. (h) Modes of education delivery--Twenty papers (19.61%) focused on online learning, eleven (10.78%) on traditional classroom-based learning, eight (7.84%) on hybrid or blended learning, but 80 (78.43%) did not specify it.

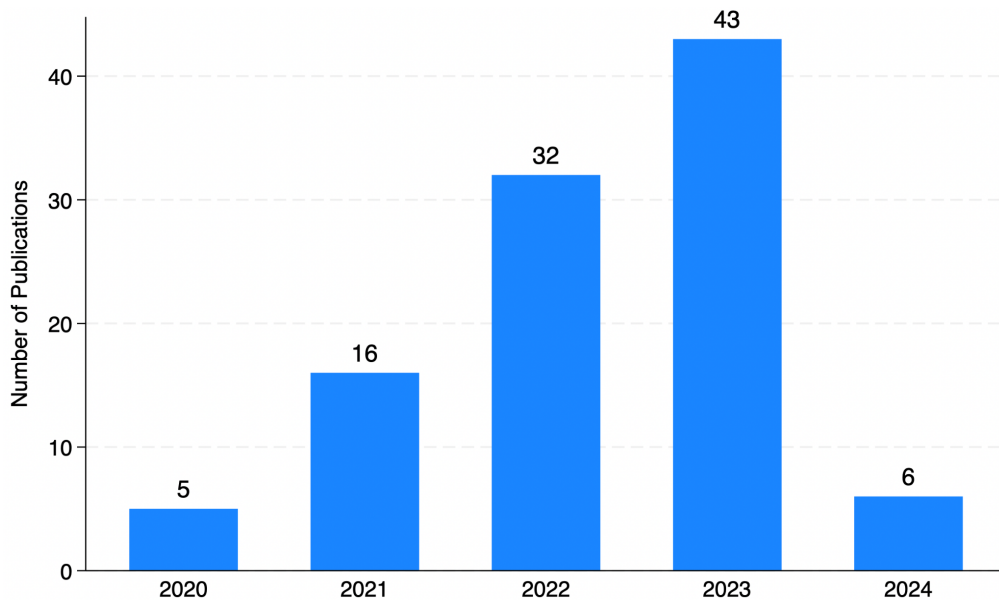


Fig. 2. Publication development of the 102 reviews.

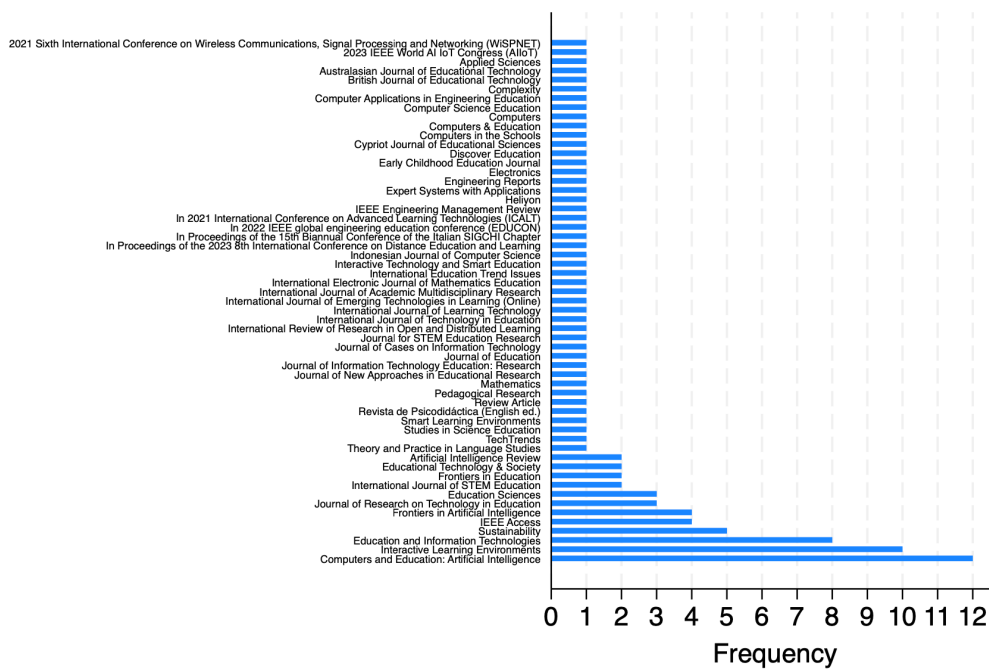


Fig. 3. Frequency of journals.

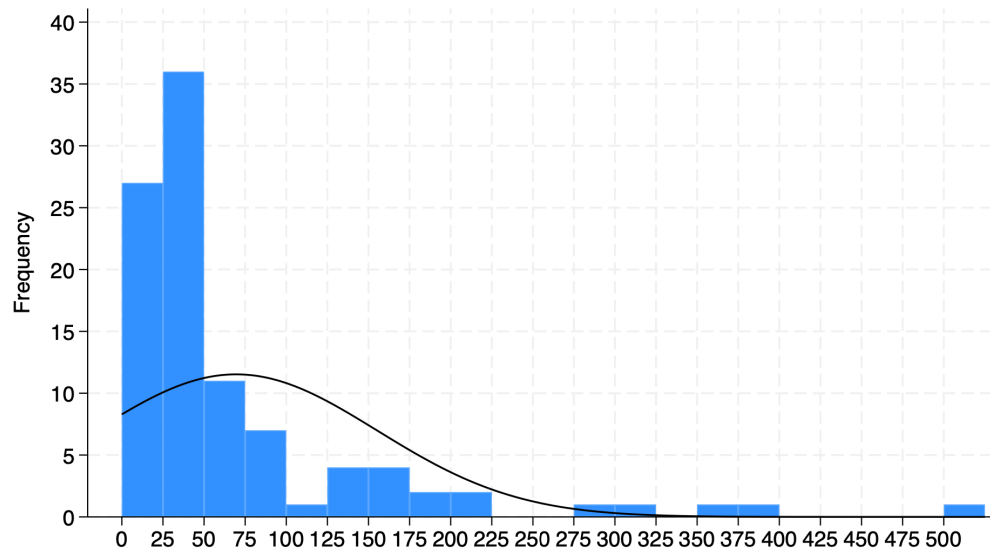


Fig. 4. Number of studies finally included.

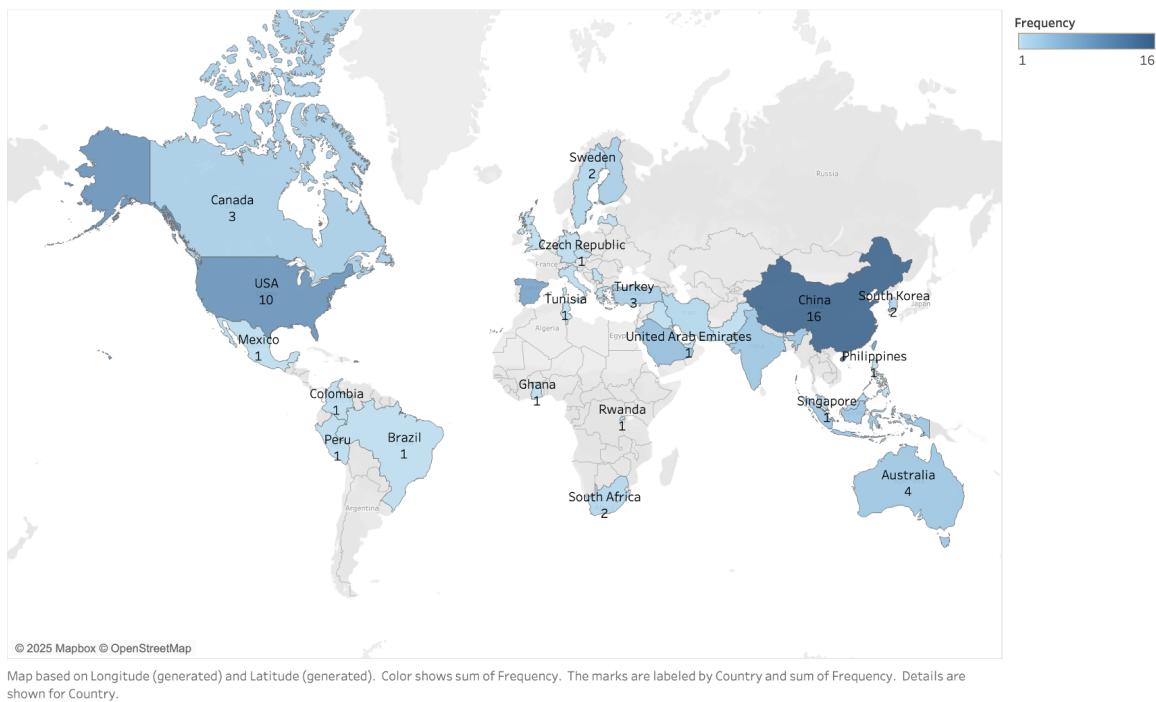


Fig. 5. Countries/regions of first authors.

3.2. RQ2: The Review Focuses

Based on the coding of review focuses, we developed a framework to organize the focus of the current reviews in three levels from the left to the right with more and more details (Fig. 6), with definitions shown in Table 3: (a) At level 1, the AIED reviews are classified as user-related and developer-related. (b) At level 2, the user-related reviews are further divided into AI Education/Literacy and AI Applications. The developer-related reviews include AI Technology Development and AIED Theories. The AI challenges were considered from a user-related perspective, a developer-related perspective, or both. (c) At level 3, AI Education and Literacy are further divided into AI Education, AI Literacy, and AI Ethics. The AI Applications are further divided into AI Functions and AI Utilization Process. The AI Technology Development is further divided into AI Techniques, AI Programming Languages, and AI Operating Platforms. This hierarchical framework provides a comprehensive structure for organizing and understanding recent advancements and discussions in AIED.

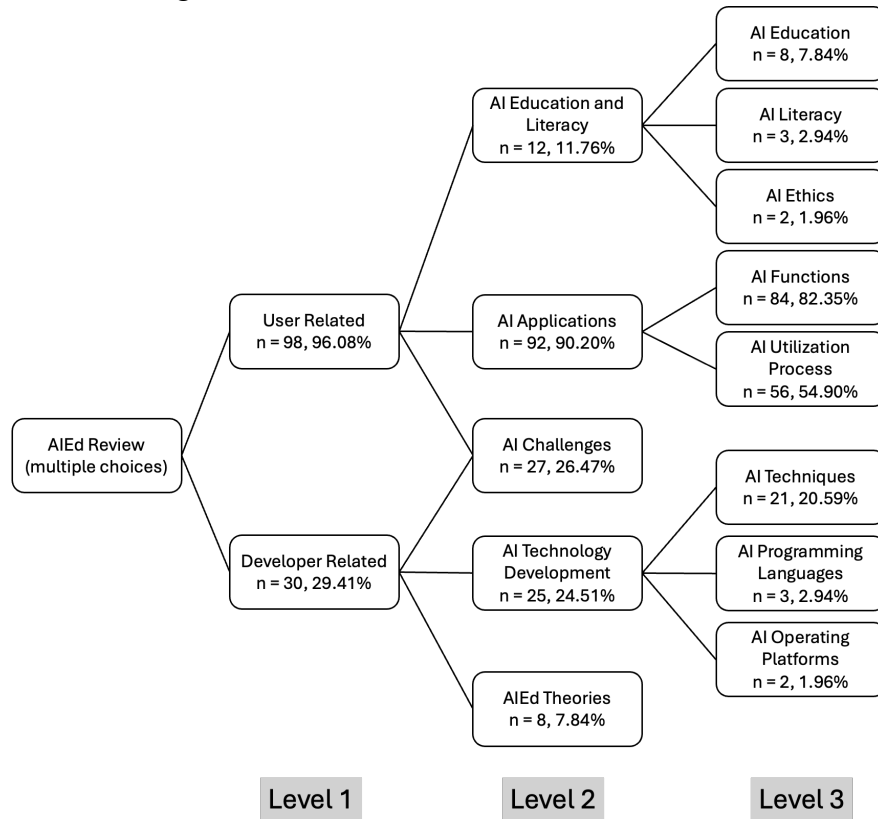


Fig. 6. The AIED review framework.

Note. As many AIED reviews include more than one focus (e.g., a review may include both AI ethics and AI applications), the sum of lower-level categories does not always equal the total for the corresponding upper-level categories.

Table 3

AIEd framework definitions.

	Themes	Definition
Level 1	User-related Theme	Focus on the experiences and needs of educators and students.
	Developer-related Themes	Focus on the technological and theoretical foundations of AI development.
Level 2	AI Education and Literacy	The development of knowledge, skills, and ethical understanding of AI in education.
	AI Applications	The use of AI tools to enhance teaching and learning.
	AI Challenges	Challenges facing both AI use and development, including technological limitations, pedagogical challenges, ethical risks, and systemic barriers.
	AI Technology Development	The research and development of foundational components that support the creation, deployment, and optimization of AI systems.
	AIEd Theories	Theories, frameworks, or principles that underpin the design and use of AIEd, potentially integrating pedagogical, psychological, or technological insights.
Level 3	AI Education	Courses, programs, or curricula focused on AI concepts and applications.
	AI literacy	The ability to understand, use, create, and evaluate AI technologies.
	AI Ethics	A set of principles and values that guide the responsible design, use, and societal impact of AI.
	AI Functions	The capabilities and tasks performed by AI systems, such as prediction, detection, classification, decision-making, and automated assessment in various applications.
	AI Utilization Process	The implementation of AI products, their impact, user attitudes, and how users interpret and act upon AI-generated outputs.
	AI Techniques	The review, analysis, and evaluation of AI approaches and algorithms, focusing on the types, characteristics, and performance of different AI technologies.
	AI Programming Languages	The coding languages used to develop and implement AI systems, such as Python, R, and Java.
	AI Operating Platforms	The environments where AI systems are deployed and run, including mobile, desktop, and web platforms.

3.2.1. User-Related Themes

The user-related reviews included AI Education and Literacy and AI Application. The AI Education and Literacy reviews ($n = 12$, 11.76%) comprised AI education ($n = 8$, 7.84%), AI literacy ($n = 3$, 2.94%), and AI ethics ($n = 2$, 1.96%). The AI Application reviews ($n = 92$, 90.20%) included AI functions ($n = 84$, 82.35%) and AI utilization process ($n = 56$, 54.90%).

3.2.1.1. AI Education and Literacy

AI literacy. AI literacy, as a key educational objective, encompasses the ability to know and understand AI, use and apply AI, evaluate and create AI, and AI ethics (Ng et al., 2021). Most papers only synthesize students' AI literacy, but few systematic reviews in our dataset synthesize teachers' AI literacy. Su et al. (2024) analyzed the learning effects of K-12 AI education using the AI literacy framework. Their findings indicate that most studies emphasize enhancing students' AI knowledge, such as fundamental AI concepts and reasoning, while fewer studies address application, creation, and ethical considerations. They suggest that future AI education should expand beyond basic knowledge to include hands-on AI creation and ethical awareness to develop AI literacy. Casal-Otero et al. (2023) found that AI literacy in K-12 education is still developing but lacks standardized frameworks, rigorous assessment strategies, and interdisciplinary integration, and most studies focus on technical skills rather than ethical and societal impacts, with a significant gap in teacher training. Similarly, Su et al. (2023) found that AI literacy education enhances young children's AI knowledge, inquiry skills, and digital awareness but lacks standardized assessment tools and teacher training. They recommended establishing performance-based measurements, structured curriculum frameworks, and professional development programs to improve AI literacy education.

AI Education. Across the literature, AI education is recognized as courses, programs, or curricula designed to introduce students or teachers to AI concepts and applications. Successful AI education can foster foundational AI literacy and problem-solving skills (Su et al., 2024). Research in AI education explores the following dimensions: (a) Pedagogical Approaches: AI education integrates interdisciplinary instructional strategies, including collaborative learning, inquiry-based, project-based, and game-based learning, grounded in constructivist learning theories (Su et al., 2024; Yue et al., 2022; Oyelere et al., 2022). While some schools effectively incorporate AI into existing curricula, others struggle due to a lack of standardized instructional methods and teacher preparedness (Su et al., 2024; Sanusi et al., 2023). (b) Learning Outcomes: AI education research examines both cognitive and non-cognitive outcomes. Cognitive aspects include AI comprehension, problem-solving skills, and creativity (Su et al., 2024; Su et al., 2022). Non-cognitive outcomes, such as attitudes, dispositions, and motivation toward AI learning, as well as ethical awareness of AI (Su et al., 2022). (c) Assessment Methods: Current AI assessment strategies primarily rely on questionnaires, assessments, and surveys to evaluate students' AI knowledge, attitudes, and motivation. These methods, including pre- and post-

assessments, structured activities, and self-reported questionnaires, vary across studies and lack standardized measurement instruments (Su et al., 2024). (d) Regional Implementation: AI education policies and practices vary globally. Countries such as China, the U.S., and Singapore have structured, government-backed AI curricula integrated into national STEM initiatives (Yue et al., 2022). India has made significant progress in AI education through policy-driven initiatives like National Education Policy 2020 and Central Board of Secondary Education's AI curriculum. However, AI education remains in its early stages, with challenges such as teacher training, infrastructure, and AI literacy still being addressed. (Karan & Angadi, 2023). In contrast, AI education in Africa faces significant challenges, including infrastructure limitations, insufficient teacher training, and the lack of government-endorsed AI curricula. While some nations, like Mauritius and Kenya, have introduced AI strategies, broader implementation remains uneven (Oyelere et al., 2022).

AI Ethics. AI ethics refers to a set of principles and values that guide the responsible design, use, and societal impact of AI. It encompasses several critical considerations: (a) Fairness and Equity: Disparities in access to AI technologies can widen the digital divide and further disadvantage underprivileged students (Crompton et al., 2024). (b) Transparency and Explainability: AI-driven educational tools often lack clear explanations of their decision-making processes (Vargas-Murillo et al., 2023), making it difficult for students and teachers to trust automated grading or personalized learning recommendations (Saputra et al., 2023). Research highlights that teachers struggle to interpret and justify AI-generated feedback, emphasizing the need for more interpretable AI models (Crompton et al., 2024). (c) Academic Integrity and Plagiarism: The rise of AI-assisted learning tools like ChatGPT presents ethical dilemmas regarding student originality, as AI-generated content is increasingly used in assignments without proper citation or acknowledgment. (Vargas-Murillo et al., 2023). The difficulty of distinguishing between human- and AI-generated work necessitates the development of academic integrity policies that incorporate AI detection tools and ethical guidelines (Saputra et al., 2023). (d) Data Privacy and Security: AI-based learning platforms collect extensive students or teachers data, leading to concerns regarding data ownership, security, and potential misuse. For example, personal information can be exploited through facial recognition and recommender systems (Saputra et al., 2023). Despite these concerns, many studies do not explicitly address ethical procedures in their methodologies, leaving potential ethical vulnerabilities unexamined (Salas-Pilco et al., 2022). (e) Human Oversight and Autonomy: While AI can enhance efficiency in grading and personalized instruction, over-reliance on automation risks diminishing the role of educators and students in critical decision-making. Research suggests that AI should complement rather than replace human judgment, and avoid students' abuse of AI tools like ChatGPT (Vargas-Murillo et al., 2023).

3.2.1.2. AI Application

AI Function. AI functions encompass the capabilities and tasks performed by AI systems, such as prediction, detection, classification, decision-making, and automated assessment in various applications. It can be categorized into six key areas in education: (a) **Teacher Support:** AI assists educators by developing curriculum materials, generating educational content, facilitating engaging teaching experiences, supporting instructional decision-making, and enhancing professional development opportunities (Tang et al., 2023; Chng et al., 2023; Crompton et al., 2024; Xu & Ouyang, 2022b). (b) **Personalized Learning and Student Engagement:** AI enables adaptive learning by tailoring instruction to students' needs, providing individualized learning paths, and enhancing engagement through interactive and motivation-driven interventions (Tang et al., 2023; Liang et al., 2023; Xu & Ouyang, 2022a). (c) **Interaction and Collaboration:** AI acts as a tutor, tutee, or peer to enhance student-student and teacher-student interactions, supports collaborative learning, and integrates with AR/VR technologies to facilitate immersive experiences and remote laboratory simulations (Jeon et al., 2023; Suryanti et al., 2024; Xu & Ouyang, 2022b; Okonkwo & Ade-Ibijola, 2021). (d) **Prediction and Monitoring:** AI predicts student and teacher performance by analyzing learning patterns and behavioral data while also monitoring students' engagement, emotions, and progress to enable early interventions and personalized support (Chng et al., 2023; Zhou et al., 2024). (e) **Assessment and Feedback:** AI automates assessment processes by generating quizzes, grading tests, detecting cheating or plagiarism, and providing formative feedback to support deeper understanding and self-assessment (Martínez-Comesaña et al., 2023; Jeon et al., 2023; Crompton et al., 2024). (f) **AI-Driven Content Generation and Knowledge Management:** AI contributes to content creation by summarizing texts, generating exercises, curating learning materials, and supporting knowledge organization, retrieval, and contextual recommendations for both students and educators (Suryanti et al., 2024; Tang et al., 2023; Xu & Ouyang, 2022b).

AI Utilization Process. AI utilization process refers to the process of the implementation of AI products in education, which includes basic elements like users, interventions, and implementation durations. The AI utilization process must address the impact of AI on users, including their attitudes and how they interpret and respond to AI-generated outputs, whereas AI functions may only be discussed in terms of theoretical frameworks or model testing. The AI utilization process can be synthesized into the following aspects: (a) **Defined users, structured interventions, and varying implementation durations:** Users of AI in K-12 education primarily involve students, teachers, and sometimes parents, with tools tailored to their specific roles (Zhang et al., 2024; Neumann, 2020). Interventions vary widely, ranging from automated feedback systems to intelligent tutoring systems (Liang et al., 2023; Hopcan et al., 2023). Implementation durations range from short-term interventions (e.g., less than one week, or four

weeks' tails) to longer-term deployments (e.g., several months, or academic semesters) (Jeho et al., 2024; Zhang et al., 2024). Wu and Yu (2022) found that short-term Chatbot interventions had a stronger effect on students' learning outcomes than long-term interventions, probably due to novelty effects of AI chatbots in a short term. (b) The Impact of AI in Balancing Benefits and Challenges: AI-driven automation in assessment reduces teacher workload, enabling educators to focus on instructional strategies and student engagement (Lo, 2023; Zhang & Aslan, 2021; González-Calatayud et al., 2021; Chiu et al., 2023). AI-powered personalized learning enhances engagement and academic performance by adapting content and pacing to individual student needs (Chiu et al., 2023; Salas-Pilco et al., 2022; García-Martínez et al., 2023). However, excessive reliance on AI-enhanced tools may limit social interactions, diminish critical thinking and problem-solving skills, and impose a cognitive load that distracts students and hinders their focus on the learning process (Lo, 2023; Okonkwo & Ade-Ibijola, 2021, Liang et al., 2021, Xu and Ouyang, 2022b). (c) Users' attitudes towards AI: Deng and Yu (2022) synthesized users' attitudes toward AI in K-12 education into two aspects—students' attitudes and teachers' attitudes. Students appreciate AI's ability to enhance learning efficiency and engagement but express concerns about overreliance and ethical implications. Teachers recognize AI's instructional benefits but highlight challenges such as student dependence and the need for adequate teacher training and support for effective implementation. (d) Interpretation and Actions: Researchers studied how users interpreted and took actions when they used AI tools. Maita et al. (2024) found that some individuals rely on AI as a primary source without concerns about plagiarism and academic integrity. However, it is important to critically evaluate AI's accuracy before taking actions. For example, Salas-Pilco et al. (2022) found that some AI algorithms may not capture contextual factors related to minority students. Their findings highlight the importance of critical evaluation before acting on AI-generated outputs. Ji et al., (2023) emphasized the need to strengthen partnerships between human language teachers and conversational AI, as well as to assess human-computer interactions in research contexts that better align with real-world language classrooms.

3.2.2. AI Challenges

The AI challenges ($n = 27$, 26.47%) were placed in the middle of the framework, as many studies considered AI challenges from either a user-related or developer-related perspective, or both. For example, from a user-related perspective, Tissenbaum and Slotta (2019) reported that teachers found AIED tools to be distracting, which prevented students from focusing on learning objectives. From a developer-related perspective, Chen et al. (2016) highlighted challenges in designing AI tools tailored for language learning, where retrieved example sentences were too difficult for lower or intermediate-level language learners, requiring developers to refine algorithms for better personalization. Similarly, Matsuda et al. (2020) reported that AI agents had

limited ability to provide meaningful feedback for free-style input, which emphasized the significant challenges for developers in building adaptable and context-aware systems.

The AI Challenges theme includes technological limitations, pedagogical hurdles, ethical threats, and systemic barriers: (a) Technological limitations hinder AI adoption as teachers often struggle to adapt AI tools to diverse classroom needs and align them with curriculum goals. Crompton et al. (2024) highlighted examples of teachers abandoning AI due to troubleshooting difficulties, while students' limited technological skills widen the gap between AI's potential and its practical use. (b) Pedagogical hurdles, such as insufficient teacher training and difficulties in integrating AI into existing curriculum, hinder effective implementation (Saputra et al., 2023; Zhou et al., 2024). (c) Ethical concerns threaten AI use in education, such as data security, algorithmic bias, and transparency (Saputra et al., 2023; Zhou et al., 2024). For example, AI predictive analytics can lead to biased outcomes or misuse of student data and harm fairness and equity (Crompton et al., 2024). (d) Systemic barriers, such as insufficient funding support and inadequate equipment and facilities, prevent AI adoption. Saputra et al. (2023) and Suryanti et al. (2024) identify key systemic barriers, including high implementation costs, unequal access to AI technologies, and slow curriculum reform. These issues are particularly acute in resource-limited schools, exacerbating inequities in educational opportunities.

3.2.3. Developer-Related Themes

The developer-related reviews included AI Technology Development and AIEd Theories. The AI Technology Development theme ($n = 25$, 24.51%) included AI techniques ($n = 21$, 20.59%), AI programming languages ($n = 3$, 2.94%), and AI operating platforms ($n = 2$, 1.96%). The AIEd Theories ($n = 8$, 7.84%) provided theoretical frameworks and models supporting AIEd.

3.2.3.1. AI Technology Development

AI Techniques. AI Techniques includes the review, analysis, and evaluation of AI approaches and algorithms, focusing on the types, characteristics, and performance of different AI technologies. The synthesis of AI techniques can be categorized into two main areas (Tan et al., 2022; Chen et al., 2021; Celik et al., 2022; Luan & Tsai, 2021; Lin et al., 2023; Orji & Vassileva, 2022): (a) General AI Approaches, which include machine learning (including deep learning techniques such as convolutional neural networks and artificial neural networks), natural language processing, and reinforcement learning. (b) Specific AI Models or Algorithms, which include supervised learning (e.g. ridge regression, lasso regression, linear regression, random forest, decision tree, naive Bayes, support vector machines, k-nearest neighbors), and unsupervised learning (e.g. k-means clustering, principal component analysis, hierarchical

clustering). Three main validation approaches are used to evaluate a machine learning model (Zhai et al., 2020): (a) Self-Validation: It uses the same dataset for training and testing. (b) Split-Validation: It splits data into separate training and testing sets. (c) Cross-Validation: It uses multiple subsets of data to iteratively train and test the model.

The AI technique can be evaluated for different tasks (Tan et al., 2024; Zhai et al., 2020; Luan & Tsai, 2021; e Silva et al., 2023). Some examples are as follows. First, for classification tasks, to what extent the machine classifications are consistent with human consent classifications (Fig. 7): (a) Accuracy: The percentage of correctly labeled (true positive) and correctly unlabeled cases (true negative) among all the cases – $(A + D)/(A + B + C + D)$. (b) Precision: The ratio of correctly labeled cases (true positive) to the total labeled cases (true positive and false positive) – $A/(A + B)$. (c) Recall (Sensitivity): The percentage of correctly labeled cases (true positive) among all the cases should be labeled (true positive and false negative) – $A/(A + C)$. (d) Specificity: The percentage of unlabeled cases (true negative) out of the total number of cases that should not be labeled (false positive and true negative) – $D/(B + D)$. (e) F1 Score: A weighted measure of precision and recall – $2 * \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$. (f) Area Under the Receiver Operating Characteristic Curve: Measures classification performance across different decision thresholds. (g) Cohen's Kappa: Measures the inter-rater agreement between human and machine scores, accounting for chance agreement. (h) Quadratic-Weighted Cohen's Kappa: Assigns different weights to disagreements based on their severity.

Human Consent Classification

Machine Classification	Human Consent Classification	
	1	0
	1	0
	A (true positive)	B (false positive) Type I Error
	C (false negative) Type II Error	D (true negative)

Fig. 7. Confusion matrix.

Note. Adapted from *Parameters for machine learning* by X. Zhai, 2019, *Internal Document: unpublished* (<https://doi.org/10.13140/RG.2.2.33563.31525>). Adapted with permission.

Second, for regression tasks—use ML to predict numeric outcomes (Tan et al., 2024): (a) Root Mean Square Error: Average difference between predicted values and actual values. (b) Mean Square Error: Average of the squared differences between predicted and actual values. (c) Mean Absolute Error: Average of the absolute differences between predicted and actual values. (d) Coefficient of Determination (R^2): Proportion of variance in the dependent variable explained by the model.

AI Programming Languages. AI programming languages refers to the coding languages used to develop and implement AI systems. Several studies, including Alalawi et al. (2023), Salas-Pilco et al. (2022), and AbdulMuhsen AlHarbi (2022), highlight Python, R, and Java as the most commonly used languages in AI-driven education research. Python is the most widely used, favored for its extensive machine learning libraries and its flexibility in AI development. R is particularly valued for statistical computing and data visualization, making it a key tool in AI-driven analysis. Java, though less commonly used for AI, remains relevant for its robustness and platform independence.

AI Operating Platforms. AI Operating Platforms refers to the environments where AI systems are deployed and run. The following operating platforms have been reviewed (Kuhail et al., 2023, Jeon et al., 2024): (a) Web-based platforms are the most widely used, as they are operating system-independent and accessible on multiple devices. (b) Mobile-based platforms, including smartphones and tablets, are gaining prominence due to their portability and real-time accessibility for learning applications. (c) Desktop-based platforms, though less common, remain relevant in structured laboratory settings where desktop computers are used for controlled chatbot interactions. (d) Smart speaker platforms, such as Amazon Alexa and Google Assistant, provide hands-free, voice-activated AI interactions, making them particularly useful in language learning. (e) immersive platforms leveraging VR/AR technologies, panoramic screens, and wearable devices are being explored for their potential to enhance user engagement and create realistic learning environments.

3.2.3.2. AIEd Theories

AIEd Theories. AIEd Theories refers to theories, frameworks, or principles that underpin the design and use of AIEd, potentially integrating pedagogical, psychological, or technological insights. The theories from the AI in K-12 education reviews were synthesized as the following aspects: (a) Cognitive and Learning Theories: Cognitive Load Theory has been used to examine how pedagogical agents (PAs) induce different types of cognitive load on learners' working memory (Sikström et al., 2022). For example, over-reliance on AI-generated content (e.g. ChatGPT) may increase extraneous cognitive load (Maita et al., 2024). Self-Regulated Learning Theory has been applied to AI-powered assessments and simulations to

analyze students' independent learning behaviors (Dai & Ke, 2022; Sikström et al., 2022). Social Cognitive Theory has been used to study how AI influences learning behaviors through observation, imitation, and interaction (Sikström et al., 2022; Maita et al., 2024). Social Cognitive Theory is the common guiding theory of the studies on virtual agents, and assessments agents (Dai & Ke, 2022). (b) Social and Interaction Theories: Social Agency Theory has guided research on PAs, suggesting that social cues can evoke learners' social schemata and enhance their learning through understanding PAs' communication (Sikström et al., 2022). The Cognitive-Affective-Social Theory of Learning in Digital Environments posits that social cues trigger parasocial, motivational, and emotional responses that affect learning (Sikström et al., 2022). Instructional Communication Theory has been explored in AI-based conversational systems to examine how communication strategies influence student-agent interactions (Sikström et al., 2022). Collaborative Learning has been integrated into AI-powered chatbots (e.g. an animated conversational agent) to improve students' critical thinking and argumentation (Kuhail et al., 2023; Dai & Ke, 2022). (c) Assessment and Feedback Theories: Constraint-Based Modeling, Item Response Theory, and Evidence-Centered Design have been referenced in AI-based assessments but remain underrepresented in empirical studies (González-Calatayud et al., 2021). The frameworks of Taxonomy of Error Types and CALF (syntactic complexity, accuracy, lexical complexity, and fluency) Measures have been used to evaluate AI's linguistic outputs and accuracy in machine-translation-assisted language learning (Deng & Yu, 2022). (d) Technology Adoption and Integration Theories: The Technology Acceptance Model has been applied to analyze AI adoption in education, involving three key components as perceived usefulness, perceived ease of use, and behavioral intention (Deng & Yu, 2022). The Technological Pedagogical Content Knowledge framework has been used to study how AI tools (e.g. ChatGPT) supports educators in transforming subject matter expertise into effective teaching practices (Maita et al., 2024; Deng & Yu, 2022). (e) Constructivist Perspectives: Vygotsky's Sociocultural Theory emphasizes the role of social robots in scaffolding individual's language and literacy learning within their Zone of Proximal Development by providing adaptive support, feedback, and interactive engagement to facilitate knowledge construction (Neumann, 2020; Kuhail et al., 2023). Constructivist theory posits that individuals actively create knowledge by integrating new information with their experiences and prior knowledge, shaping their own understanding and learning process (Maita et al., 2024). Self-Determination Theory has been used to examine AI's effects on learner autonomy and intrinsic motivation (Maita et al., 2024).

3.3. RQ3. The Quality Assessment Results

As shown in Table 4, our quality assessment revealed that most review studies excelled in most dimensions: (a) *Rationale*: The majority of systematic reviews ($n = 86$, 84.31%) clearly stated the importance of the topic and identified gaps in prior research. (b) *Objectives*: Nearly all

papers ($n = 101$, 99.02%) clearly addressed their objectives. (c) *Searching Databases*: Most reviews ($n = 98$, 96.08%) explicitly described the intended information sources. (d) *Searching Strategy*: Ninety-one studies (89.22%) clearly presented their search strategies, ensuring reproducibility. (e) *Inclusion and Exclusion Criteria*: Most reviews ($n = 90$, 88.24%) clearly specified inclusion and exclusion criteria based on study characteristics. (f) *Screening Process*: Most papers ($n = 78$, 76.47%) detailed the screening processes used for each phase of the review. (g) *Coding Rubrics*: A majority of papers ($n = 72$, 70.59%) clearly listed and defined all variables for coding literature to answer research questions or achieve research goals. (h) *Presentations of Findings*: Most studies ($n = 85$, 83.33%) clearly presented findings based on the research questions, demonstrating good organization and specificity. (i) *Results Supported by Data*: Most studies ($n = 85$, 83.33%) showed strong evidence supporting their results.

However, most reviews need improvement in four dimensions: (a) *Data Management*: The majority of reviews ($n = 76$, 74.51%) did not describe data management mechanisms or software. (b) *Extraction Process*: Thirty-seven papers (36.27%) did not describe their data extraction methods, and 17 (16.67%) provided only somewhat clear explanations. Many researchers confuse the screening process with the extraction process, often clearly stating the screening process but overlooking the extraction process. (c) *Analysis Methods*: Fewer than two-thirds of reviews ($n = 58$, 56.86%) clearly described their analysis methods. (d) *Limitations*: Over 39% of studies either did not state ($n = 31$, 30.39%) or only somewhat described ($n = 9$, 8.82%) their limitations.

Table 4

Quality assessment results.

Items	Category	Frequency	Percent	Cumulative Percentage
Rationale	Not stated	1	0.98	0.98
	Somewhat stated	15	14.71	15.69
	Clearly stated	86	84.31	100
Objectives	Not stated	0	0	0
	Somewhat stated	1	0.98	0.98
	Clearly stated	101	99.02	100
Databases	Not stated	0	0	0
	Somewhat stated	4	3.92	3.92
	Clearly stated	98	96.08	100
Search	Not stated	1	0.98	0.98
Strategy	Somewhat stated	10	9.8	10.78
	Clearly stated	91	89.22	100
Criteria	Not stated	1	0.98	0.98

Management	Somewhat stated	11	10.78	12.75
	Clearly stated	90	88.24	100
	Not stated	76	74.51	74.51
Screening	Somewhat stated	0	0	74.51
	Clearly stated	26	25.49	100
	Not stated	4	3.92	3.92
Extracting	Somewhat stated	20	19.61	23.53
	Clearly stated	78	76.47	100
	Not stated	37	36.27	36.27
Coding	Somewhat stated	17	16.67	52.94
	Clearly stated	48	47.06	100
	Not stated	10	9.8	9.8
Methods	Somewhat stated	20	19.61	29.41
	Clearly stated	72	70.59	100
	Not stated	32	31.37	31.37
Presentation	Somewhat stated	12	11.76	43.14
	Clearly stated	58	56.86	100
	Not appropriate	3	2.94	2.94
Results	Somewhat appropriate	21	20.59	23.53
	Highly appropriate	78	76.47	100
	Not at all	1	0.98	0.98
Limitations	Somewhat	16	15.69	16.67
	Strongly	85	83.33	100
	Not stated	31	30.39	30.39
	Somewhat stated	9	8.82	39.21
	Clearly stated	62	60.78	100

Note. Percentages are rounded to two decimal places. Total percentages may differ slightly due to rounding adjustments. The total number of papers is $n = 102$.

4. Discussion

4.1. RQ1: Characteristics of the Studies

The analysis of study characteristics reveals key trends shaping the landscape of AI reviews in K-12 education. The rapid rise in reviews from 2020 to 2023 highlights growing

interest in synthesizing AI research, driven by advancements in technology. Although we searched for AIED reviews without a time limit, all the 102 systematic reviews that met our criterion were published after 2020, suggesting a turning point when empirical studies on AI in education reached a critical mass for comprehensive synthesis from 2020. This observation marks a new era in the field and aligns with findings by other researchers (e.g., Wang and Cheng, 2021, Martin et al., 2024, and Marques, 2020), who reported a surge in AI-related publications after 2018. Additionally, the broad journal distribution across 57 venues underscores interdisciplinary interest. The key journals, such as *Computers and Education: Artificial Intelligence* ($n = 12$) and *Interactive Learning Environments* ($n = 10$), *Education and Information Technologies* ($n = 8$), consolidate AIED knowledge and foster a cohesive understanding of AIED.

The funding landscape shows that nearly 40% of the reviews received financial support, reflecting the recognition of AIED importance in K-12 education. Most first authors are from China ($n = 16$, 15.69%), Hong Kong ($n = 11$, 10.78%), and USA ($n = 10$, 9.80%), which reflects the strong investments in AI research in these regions. The underrepresentation of elsewhere underscores the disparity among different regions and importance of global collaboration and inclusivity. The lack of reviews in other regions might be also due to the fact that we only included the articles published in English.

More than half of the reviews (57.84%) only included empirical studies on AIED. The emphasis on empirical studies highlights the demand for evidence-based insights. Nearly 30% reviews did not clarify whether only empirical studies are included, which highlights the importance of clarifying inclusion criteria. Lastly, some reviews did not specify educational settings and delivery modes where AI was applied, which limits the practical applicability of the findings.

Future review studies should clarify contextual factors in the reviewed studies, such as educational settings and delivery modes, to enhance the practical applicability of findings.

4.2. RQ2: The Review Focuses

To answer RQ2 we created a framework (Fig. 6) to highlight both the diversity and interconnectedness of the research themes, illustrating the multifaceted nature of AIED reviews. We categorized the systematic reviews into two primary dimensions of focus in the integration of AI into K-12 education: user-related themes, which focus on the experiences and needs of educators and students, and developer-related themes, which primarily explore the technological and theoretical foundations of AI systems. Using both quantitative and qualitative methods, we

summarize the major findings of the reviews in the results section. This section discusses the focuses and suggests future research directions.

4.2.1. User-Related Themes

Most AIED reviews focused on AI applications (90.20%) while much fewer reviewers focus on AI education and literacy (11.76%), which highlights a predominant focus on leveraging AI technologies for educational purposes rather than equipping educators and students with the knowledge to understand and navigate these tools. The prioritization of functionality over foundational preparedness may hinder users' ability to adopt and manage AI tools effectively, leading to risks such as misuse, over-reliance, or unintended consequences like reinforcing biases or inequities. Moreover, the focus on applications may overshadow the importance of teacher-student relationships and human-centered approaches to learning.

Within the application, the large number of reviews focused on AI Functions (82.35%). These reviews discussed the possible functions of AI tools. The large number of AI function reviews may be attributed to the rapid development of AI solutions tailored for teaching and learning. In contrast, the AI Utilization Process (54.90%), which synthesizes the empirical studies that implement AI tools in educational settings, was less prevalent than AI Functions. The disparity between AI function and AI utilization suggests that more AI functions receive significant attention due to their emphasis on the technical capabilities of AI systems, such as prediction, classification, and decision-making; fewer studies reviewed the empirical studies of utilizing AI tools, such as how users perceive and use AI tools and how AI tools impact learning and teaching. It is understandable that it takes time for new technology to be implemented and examined. More studies and reviews need to address AI implementation in real classrooms, such as user attitudes toward AI, interpretation of AI output, and its impact on learners and teachers.

The limited focus on AI education (7.84%), AI literacy (2.94%), and AI ethics (1.96%) within AI Education and Literacy suggest that integrating AI-related content into K-12 curricula is still in its early stages. The underrepresentation of AI ethics is particularly concerning, as it reflects insufficient attention to critical issues such as data privacy, algorithmic bias, and the broader societal implications of AI. If the ethical concerns are not sufficiently studied and addressed, AI tools may be misused by some and rejected by others, leading to fairness and equity issues. Therefore, more attention to AI education, literacy, and ethics in future research could help bridge this gap by better preparing students and educators to understand and use AI tools while critically evaluating their implications, promoting ethical and effective AI adoption in education.

4.2.2. AI Challenges

Challenges are discussed in both user- and developer-related reviews. Those reviews have highlighted four major challenges: technological limitations, pedagogical hurdles, ethical threats, and systemic barriers. To address these challenges, (a) well-structured teacher training programs should be developed to incorporate AI literacy and emphasize both technical and ethical competencies. (b) Studies should explore models of AI implementation that accommodate diverse educational contexts and promote accessible and equitable resource allocation. (c) Policymakers and educational leaders should invest in infrastructure and improve fair distribution of resources, particularly in underserved communities. (d) Longitudinal studies should be conducted to track the impact of AI-supported learning tools over time and assess their effectiveness in bridging technological and pedagogical gaps. (e) Interdisciplinary approaches should be emphasized, integrating insights from education, policy, and technology to develop inclusive AI practices that benefit all learners.

4.2.3. Developer-Related Themes

Compared to user-related reviews (96.08%), developer-related reviews (29.41%) are significantly less represented. Within these developer-focused reviews, the emphasis on AI technology development (24.51%) and AIED theories (7.84%) suggests an awareness of AI's technical and conceptual foundations. However, these areas have not received sufficient attention, highlighting a need for further exploration. One possible reason for this imbalance is the inclusion and exclusion criteria of our umbrella review, which excluded purely technology-based reviews—those focusing solely on AI algorithms without discussing their educational applications or implications. As a result, research focus on AI development may be underrepresented in the dataset. Additionally, research on AI in K-12 education often prioritizes user-centered perspectives, such as how AI tools enhance learning and teaching, over investigations into the design. Future research should encourage interdisciplinary collaborations between AI developers and education researchers to bridge the gap between technical innovation and practical application.

Within AI technology development, AI techniques (20.59%) encompass general approaches (e.g. machine learning), specific algorithms (e.g., decision tree), and their performance evaluations. While most reviews on AI techniques discuss the types of AI techniques, fewer of them examine their evaluation metrics or systematically review the evaluation metrics. This difference indicates that while synthesizing AI technique types is relatively common, more attention should be given to evaluate their performance in educational contexts.

Compared to AI techniques (20.59%), our umbrella review shows a lower percentage of studies focusing on AI programming languages (2.94%) and operating platforms (1.96%). One

possible reason is that programming languages and platforms, while crucial for AI development, are viewed as developer-centric concerns and are less relevant to education researchers who focus on how AI tools support learning and teaching. In contrast, AI techniques and evaluation metrics are more directly tied to the functionality and effectiveness of AI in education, making them more relevant for educational research and synthesis. Future studies could explore whether certain programming environments or platforms influence the adaptability of AI-driven educational technologies. While these aspects may be less central to AIED research, understanding their potential implications could help inform the development of AI tools that are more compatible with diverse educational contexts.

AIED theories (7.84%) remain an underdeveloped area despite their critical role in providing conceptual frameworks for designing and evaluating AI systems in education. This limited focus may arise from the challenge of integrating theoretical insights into real-world educational settings. Further research is needed to fill this gap by examining how theoretical frameworks inform the design and evaluation of AI tools and ensuring that technological advancements align with pedagogical goals.

4.3. RQ3. The Quality Assessment Results

The quality assessment reveals both commendable strengths and notable gaps in systematic reviews. High percentage of reviews clearly presented their rationale (84.31%), objectives (99.02%), search strategies (89.22%) and findings supported by evidence (83.33%). That is, these reviews primarily pertain to conceptual and organizational elements of systematic reviews and align closely with the initial planning and reporting phases of systematic review guidelines like PRISMA, where researchers are more likely to focus on providing clarity and structure.

On the other hand, the insufficient items—such as data management mechanisms (74.51% lacking clarity), data extraction methods (36.27% unclear or absent), analysis methods (43.14% insufficiently described), and limitations (39% inadequately addressed)—are largely related to technical and procedural elements. These aspects require more specialized knowledge, methodological rigor, and close attention to detail, which may be challenging for researchers with varied levels of expertise. The findings suggest that while many researchers excel in planning and reporting basic components of systematic reviews, they may need to specify the specialized review tools and improve procedural rigor as well as transparency.

Addressing these gaps requires targeted efforts to improve technical skills and procedural rigor among researchers. The efforts may include providing training on data management and extraction processes, encouraging the use of standardized tools, and fostering a cultural shift that values transparency and reporting limitations. In addition, journals and reviewers can help

improve the review quality by promoting review guidelines and emphasizing the importance of comprehensive and transparent reporting.

5. Conclusions and Limitations

5.1 Major findings

This umbrella review made the following major contributions: First, we generated an AIED Review framework (Fig. 6) to map the territory of AIED, guide researchers in designing their review studies, and address underexplored areas. Using this framework, reviews on AIED were categorized into user-related themes and developer-related themes. We identified that more reviews are needed to focus on AI education and literacy, AI utilization process, and AIED theories.

Second, we synthesized the knowledge in AIED reviews. This knowledge provides an overview of AIED in K-12 from multiple aspects, such as AI education, AI applications, AI technologies, and AIED theory. This synthesized knowledge can help newcomers to the AIED field to have a quick view of this fast-evolving field and draw researchers' attention to the understudied directions.

Third, we developed a quality assessment rubric to evaluate systematic reviews, which can be used for future umbrella reviews across various fields. We found that while many researchers excel in planning and reporting basic components of systematic reviews, gaps remain in data management, extraction processes, analysis methods, and reporting limitations. Future systematic reviews may improve the transparency and clarity of those components in their reporting.

5.2 Limitations and Implications

Despite our efforts to ensure a comprehensive umbrella review, several limitations should be acknowledged. First, although we searched across seven major academic databases and conducted an updated search in 2024, our review may still have missed relevant studies published outside these sources and time frame. Future research should include a broader range of sources and continuously updated literature to capture the fast-evolving landscape.

Second, we excluded purely technology-based reviews that only studied AI algorithms without discussing their applications in K-12 education. The studies that only focus on AI

technology are beyond the scope of this umbrella review. Therefore, umbrella reviews on pure AI technology deserve their own attention.

Third, our umbrella review only focused on K–12 learning and teaching, excluding other educational settings (e.g., higher education) and other AI functions (e.g., AI's role in administration, such as enhancing management platforms and improving staff efficiency). Future umbrella reviews may synthesize the AIEd reviews in those contexts.

References

- AbdulMuhsen Alharbi, A. (2022). The uses of machine learning (ML) in teaching and learning English language: A methodical review. *The Educational Journal of the Faculty of Education at Sohag University*, 93(93), 25-52.
<https://doi.org/10.21608/edusohag.2022.212355>
- Alalawi, K., Athauda, R., & Chiong, R. (2023). Contextualizing the current state of research on the use of machine learning for student performance prediction: A systematic literature review. *Engineering Reports*, 5(12), Article e12699. <https://doi.org/10.1002/eng2.12699>
- Almelhes, S. A. (2023). A review of artificial intelligence adoption in second-language learning. *Theory & Practice in Language Studies (TPLS)*, 13(5), 1259-1269.
<https://doi.org/10.17507/tpls.1305.21>
- Aromataris, E., Fernandez, R., Godfrey, C. M., Holly, C., Khalil, H., & Tungpunkom, P. (2015). Summarizing systematic reviews: Methodological development, conduct and reporting of an umbrella review approach. *JBIM Evidence Implementation*, 13(3), 132-140.
<https://doi.org/10.1097/XEB.0000000000000055>
- Barth, J., De Boer, W. E. L., Busse, J. W., Hoving, J. L., Kedzia, S., Couban, R., Fischer, K., Von Allmen, D. Y., Spanjer, J., & Kunz, R. (2017). Inter-rater agreement in evaluation of disability: Systematic review of reproducibility studies. *BMJ (Clinical research ed.)*, 356, j14. <https://doi.org/10.1136/bmj.j14>
- Becker, L. A., & Oxman, A. D. (2008). Chapter 22: Overviews of reviews. In J. P. T. Higgins & S. Green (Eds.), *Cochrane handbook for systematic reviews of interventions* (pp.607-631). Wiley.
- Bhutoria, A. (2022). Personalized education and artificial intelligence in the United States, China, and India: A systematic review using a human-in-the-loop model. *Computers and Education: Artificial Intelligence*, 3, Article 100068.
<https://doi.org/10.1016/j.caeai.2022.100068>
- Casal-Otero, L., Catala, A., Fernández-Morante, C., Taboada, M., Cebreiro, B., & Barro, S. (2023). AI literacy in K-12: A systematic literature review. *International Journal of STEM Education*, 10(1), 29. <https://doi.org/10.1186/s40594-023-00418-7>
- Celik, I., Dindar, M., Muukkonen, H., & Järvelä, S. (2022). The promises and challenges of artificial intelligence for teachers: A systematic review of research. *TechTrends*, 66(4), 616-630. <https://doi.org/10.1007/s11528-022-00715-y>
- Chen, X., Zou, D., Cheng, G., & Xie, H. (2021, July). Artificial intelligence-assisted personalized language learning: Systematic review and co-citation analysis. In 2021

- international conference on advanced learning technologies (ICALT)* (pp. 241-245). IEEE.
<https://doi.org/10.1109/ICALT52272.2021.00079>
- Chen, H. H. J., Wu, J. C., Yang, C. T. Y., & Pan, I. (2016). Developing and evaluating a Chinese collocation retrieval tool for CFL students and teachers. *Computer Assisted Language Learning*, 29(1), 21-39. <https://doi.org/10.1080/09588221.2014.889711>
- Chen, L., Chen, P., & Lin, Z. (2020a). Artificial intelligence in education: A review. *Ieee Access*, 8, 75264-75278. <https://doi.org/10.1109/ACCESS.2020.2988510>
- Chen, X., Xie, H., Zou, D., & Hwang, G. J. (2020b). Application and theory gaps during the rise of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 1, Article 100002. <https://doi.org/10.1016/j.caeai.2020.100002>
- Chiu, T. K., Xia, Q., Zhou, X., Chai, C. S., & Cheng, M. (2023). Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 4, Article 100118. <https://doi.org/10.1016/j.caeai.2022.100118>
- Chng, E., Tan, A. L., & Tan, S. C. (2023). Examining the use of emerging technologies in schools: A review of artificial intelligence and immersive technologies in STEM education. *Journal for STEM Education Research*, 6(3), 385-407. <https://doi.org/10.1007/s41979-023-00092-y>
- Conn, V. S., & Coon Sells, T. G. (2014). WJNR welcomes umbrella reviews. *Western Journal of Nursing Research*, 36(2), 147-151. <https://doi.org/10.1177/0193945913506968>
- Crompton, H., Jones, M. V., & Burke, D. (2024). Affordances and challenges of artificial intelligence in K-12 education: A systematic review. *Journal of Research on Technology in Education*, 56(3), 248-268. <https://doi.org/10.1080/15391523.2022.2121344>
- Cuellar-Rojas, O., Hincapie-Montoya, M., Contero, M., & Guemes-Castorena, D. (2022). Bibliometric analysis and systematic literature review of the intelligent tutoring systems. *Frontiers In Education*, 7, Article 1047853. <https://doi.org/10.3389/feduc.2022.1047853>
- Dai, C.P., & Ke, F. (2022). Educational applications of artificial intelligence in simulation-based learning: A systematic mapping review. *Computers and Education: Artificial Intelligence*, 3, Article 100087. <https://doi.org/10.1016/j.caeai.2022.100087>
- Deng, X., & Yu, Z. (2022). A systematic review of machine-translation-assisted language learning for sustainable education. *Sustainability*, 14(13), 7598. <https://doi.org/10.3390/su14137598>
- García-Martínez, I., Fernández-Batanero, J. M., Fernández-Cerero, J., & León, S. P. (2023). Analysing the impact of artificial intelligence and computational sciences on student

- performance: Systematic review and meta-analysis. *Journal of New Approaches in Educational Research*, 12(1), 171-197.
<https://doi.org/10.7821/naer.2023.1.1240>
- Gennari, R., Melonio, A., Pellegrino, M. A., & D'Angelo, M. (2023). How to playfully teach AI to young learners: A systematic literature review. In *Proceedings of the 15th Biannual Conference of the Italian SIGCHI Chapter* (pp.1-9). Association for Computing Machinery.
<https://doi.org/10.1145/3605390.3605393>
- Gisev N, Bell JS, Chen TF. (2013). Interrater agreement and interrater reliability: Key concepts, approaches, and applications. *Res Social Adm Pharm*, 9(3), 330-338.
<https://doi.org/10.1016/j.sapharm.2012.04.004>
- González-Calatayud, V., Prendes-Espinosa, P., & Roig-Vila, R. (2021). Artificial intelligence for student assessment: A systematic review. *Applied sciences*, 11(12), 5467.
<https://doi.org/10.3390/app11125467>
- Gosling, C. J., Solanes, A., Fusar-Poli, P., & Radua, J. (2023). Metaumbrella: The first comprehensive suite to perform data analysis in umbrella reviews with stratification of the evidence. *BMJ Ment Health*, 26(1), 1-8. <https://doi.org/10.1136/bmjment-2022-300534>
- Grant, M. J., & Booth, A. (2009). A typology of reviews: An analysis of 14 review types and associated methodologies. *Health information & libraries journal*, 26(2), 91-108.
<https://doi.org/10.1111/j.1471-1842.2009.00848.x>
- Hopcan, S., Polat, E., Ozturk, M. E., & Ozturk, L. (2023). Artificial intelligence in special education: A systematic review. *Interactive Learning Environments*, 31(10), 7335-7353.
<https://doi.org/10.1080/10494820.2022.2067186>
- Hwang, G. J., Xie, H., Wah, B. W., & Gašević, D. (2020). Vision, challenges, roles and research issues of artificial intelligence in education. *Computers and Education: Artificial Intelligence*, 1, Article 100001. <https://doi.org/10.1016/j.caeai.2020.100001>
- Jeon, J., Lee, S., & Choi, S. (2024). A systematic review of research on speech-recognition chatbots for language learning: Implications for future directions in the era of large language models. *Interactive Learning Environments*, 32(8), 4613-4631.
<https://doi.org/10.1080/10494820.2023.2204343>
- Ji, H., Han, I., & Ko, Y. (2023). A systematic review of conversational AI in language education: Focusing on the collaboration with human teachers. *Journal of Research on Technology in Education*, 55(1), 48-63. <https://doi.org/10.1080/15391523.2022.2142873>
- Karan, B., & Angadi, G. R. (2025). Artificial intelligence integration into school education: A review of Indian and foreign perspectives. *Millennial Asia*, 16(1), 173-199.
<https://doi.org/10.1177/09763996231158229>

- Kelly, S., Kaye, S. A., & Oviedo-Trespalacios, O. (2023). What factors contribute to the acceptance of artificial intelligence? A systematic review. *Telematics and Informatics*, 77, Article 101925. <https://doi.org/10.1016/j.tele.2022.101925>
- Koo, T. K., & Li, M. Y. (2016). A guideline of selecting and reporting intraclass correlation coefficients for reliability research. *Journal of Chiropractic Medicine*, 15(2), 155-163. <https://doi.org/10.1016/j.jcm.2016.02.012>
- Kuhail, M. A., Alturki, N., Alramlawi, S., & Alhejori, K. (2023). Interacting with educational chatbots: A systematic review. *Education and Information Technologies*, 28(1), 973-1018. <https://doi.org/10.1007/s10639-022-11177-3>
- Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159-174. <https://doi.org/10.2307/2529310>
- Liang, J. C., Hwang, G. J., Chen, M. R. A., & Darmawansah, D. (2023). Roles and research foci of artificial intelligence in language education: An integrated bibliographic analysis and systematic review approach. *Interactive Learning Environments*, 31(7), 4270-4296. <https://doi.org/10.1080/10494820.2021.1958348>
- Lin, C. C., Huang, A. Y., & Lu, O. H. (2023). Artificial intelligence in intelligent tutoring systems toward sustainable education: A systematic review. *Smart Learning Environments*, 10(1), 41. <https://doi.org/10.1186/s40561-023-00260-y>
- Lo, C. K. (2023). What is the impact of ChatGPT on education? A rapid review of the literature. *Education Sciences*, 13(4), Article 410. <https://doi.org/10.3390/educsci13040410>
- Luan, H., & Tsai, C. C. (2021). A review of using machine learning approaches for precision education. *Educational Technology & Society*, 24(1), 250-266. <https://www.jstor.org/stable/26977871>
- Maita, I., Saide, S., Putri, A. M., & Muwardi, D. (2024). Pros and cons of artificial intelligence—ChatGPT adoption in education settings: A literature review and future research agendas. *IEEE Engineering Management Review*, 52(3), 27-42. <https://doi.org/10.1109/EMR.2024.3394540>
- Marques, L. S., Gresse von Wangenheim, C., & Hauck, J. C. (2020). Teaching machine learning in school: A systematic mapping of the state of the art. *Informatics in Education*, 19(2), 283-321. <https://doi.org/10.15388/infedu.2020.14>
- Martin, F., Zhuang, M., & Schaefer, D. (2024). Systematic review of research on artificial intelligence in K-12 education (2017–2022). *Computers and Education: Artificial Intelligence*, 6, 100195. <https://doi.org/10.1016/j.caeai.2023.100195>

- Martínez-Comesaña, M., Rigueira-Díaz, X., Larrañaga-Janeiro, A., Martínez-Torres, J., Ocaranza-Prado, I., & Kreibel, D. (2023). Impact of artificial intelligence on assessment methods in primary and secondary education: Systematic literature review. *Revista de Psicodidáctica (English ed.)*, 28(2), 93-103. <https://doi.org/10.1016/j.psicoe.2023.06.002>
- Matsuda, N., Weng, W., & Wall, N. (2020). The effect of metacognitive scaffolding for learning by teaching a teachable agent. *International Journal of Artificial Intelligence in Education*, 30(1), 1-37. <https://doi.org/10.1007/s40593-019-00190-2>
- McCarthy, J., Minsky, M. L., Rochester, N., & Shannon, C. E. (2006). A proposal for the dartmouth summer research project on artificial intelligence, august 31, 1955. *AI magazine*, 27(4), 12. <https://doi.org/10.1609/aimag.v27i4.1904>
- McGraw, K. O., & Wong, S. P. (1996). Forming inferences about some intraclass correlation coefficients. *Psychological methods*, 1(1), 30-46. <https://doi.org/10.1037/1082-989X.1.1.30>
- Mousavinasab, E., Zarifsanaiey, N., R. Niakan Kalhori, S., Rakhshan, M., Keikha, L., & Ghazi Saeedi, M. (2021). Intelligent tutoring systems: A systematic review of characteristics, applications, and evaluation methods. *Interactive Learning Environments*, 29(1), 142-163. <https://doi.org/10.1080/10494820.2018.1558257>
- Munn, Z., Stone, J. C., Aromataris, E., Klugar, M., Sears, K., Leonardi-Bee, J., & Barker, T. H. (2023). Assessing the risk of bias of quantitative analytical studies: Introducing the vision for critical appraisal within JBI systematic reviews. *JBI Evidence Synthesis*, 21(3), 467-471. <https://doi.org/10.11124/JBIES-22-00224>
- Negarandeh, R., & Beykmirza, R. (2020). Quality assessment in systematic reviews: The importance of choosing the right tools. *Nursing Practice Today*, 7(3), 161-162. <https://doi.org/10.18502/npt.v7i3.3342>
- Neumann, M. M. (2020). Social robots and young children's early language and literacy learning. *Early Childhood Education Journal*, 48(2), 157-170. <https://doi.org/10.1007/s10643-019-00997-7>
- Ng, D. T. K., Leung, J. K. L., Chu, S. K. W., & Qiao, M. S. (2021). Conceptualizing AI literacy: An exploratory review. *Computers and Education: Artificial Intelligence*, 2, Article 100041. doi:10.1016/j.caeai.2021.100041
- Okonkwo, C. W., & Ade-Ibijola, A. (2021). Chatbots applications in education: A systematic review. *Computers and Education: Artificial Intelligence*, 2, Article 100033. <https://doi.org/10.1016/j.caeai.2021.100033>
- Orji, F. A., & Vassileva, J. (2022). Automatic modeling of student characteristics with interaction and physiological data using machine learning: A review. *Frontiers in Artificial Intelligence*, 5, Article 1015660. <https://doi.org/10.3389/frai.2022.1015660>

- Oyelere, S. S., Sanusi, I. T., Agbo, F. J., Oyelere, A. S., Omidiora, J. O., Adewumi, A. E., & Ogbebor, C. (2022, March). Artificial intelligence in African schools: Towards a contextualized approach. In *2022 IEEE global engineering education conference (EDUCON)* (pp. 1577-1582). IEEE.
<https://doi.org/10.1109/EDUCON52537.2022.9766550>.
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., McGuinness, L.A., Stewart, L.A., Thomas, J., Tricco, A.C., Welch, V.A., Whiting, P., & Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, 372. <https://doi.org/10.1136/bmj.n71>
- Phuong, T. T., Foster, M. J., & Reio Jr, T. G. (2020). Faculty development: A systematic review of review studies. *New Horizons in Adult Education and Human Resource Development*, 32(4), 17-36. <https://doi.org/10.1002/nha3.20294>
- Ramesh, D., & Sanampudi, S. (2022). An automated essay scoring systems: A systematic literature review. *Artificial Intelligence Reviews*, 55(3), 2495-2527.
<https://doi.org/10.1007/s10462-021-10068-2>
- Salas-Pilco, S. Z., Xiao, K., & Hu, X. (2022). Artificial intelligence and learning analytics in teacher education: A systematic review. *Education Sciences*, 12(8), 569.
<https://doi.org/10.3390/educsci12080569>
- Salas-Pilco, S. Z., Xiao, K., & Oshima, J. (2022). Artificial intelligence and new technologies in inclusive education for minority students: A systematic review. *Sustainability*, 14(20), 13572. <https://doi.org/10.3390/su142013572>
- Sanusi, I. T., Oyelere, S. S., Vartiainen, H., Suhonen, J., & Tukiainen, M. (2023). A systematic review of teaching and learning machine learning in K-12 education. *Education and Information Technologies*, 28(5), 5967-5997. <https://doi.org/10.1007/s10639-022-11416-7>
- Saputra, I., Astuti, M., Sayuti, M., & Kusumastuti, D. (2023). Integration of artificial intelligence in education: Opportunities, challenges, threats and obstacles. A Literature Review. *The Indonesian Journal of Computer Science*, 12(4), 1590-1600.
<https://doi.org/10.33022/ijcs.v12i4.3266>
- Sikström, P., Valentini, C., Sivunen, A., & Kärkkäinen, T. (2022). How pedagogical agents communicate with students: A two-phase systematic review. *Computers & Education*, 188, Article 104546. <https://doi.org/10.1016/j.compedu.2022.104564>
- Silva, L. C., Sobrinho, Á. A. D. C. C., Cordeiro, T. D., Melo, R. F., Bittencourt, I. I., Marques, L. B., Matos, D. D. M. D. C., Silva, A. P. D., & Isotani, S. (2023). Applications of

- convolutional neural networks in education: A systematic literature review. *Expert Systems with Applications*, 231, Article 120621. <https://doi.org/10.1016/j.eswa.2023.120621>
- Southworth, J., Migliaccio, K., Glover, J., Glover, J. N., Reed, D., McCarty, C., Brendemuhl, J., & Thomas, A. (2023). Developing a model for AI across the curriculum: Transforming the higher education landscape via innovation in AI literacy. *Computers and Education: Artificial Intelligence*, 4, Article 100127. <https://doi.org/10.1016/j.caeai.2023.100127>
- Su, J., Guo, K., Chen, X., & Chu, S. (2024). Teaching artificial intelligence in K-12 classrooms: A scoping review. *Interactive Learning Environments*, 32(9), 5207-5226. <https://doi.org/10.1080/10494820.2023.2212706>
- Su, J., Ng, D. T. K., & Chu, S. K. W. (2023). Artificial intelligence (AI) literacy in early childhood education: The challenges and opportunities. *Computers and Education: Artificial Intelligence*, 4, Article 100124. <https://doi.org/10.1016/j.caeai.2023.100124>
- Su, J., Zhong, Y., & Ng, D. T. K. (2022). A meta-review of literature on educational approaches for teaching AI at the K-12 levels in the Asia-Pacific region. *Computers and Education: Artificial Intelligence*, 3, Article 100065. <https://doi.org/10.1016/j.caeai.2022.100065>
- Suryanti, R., Jahidin, J., & Fadlil, M. (2024). Artificial intelligence in education: Bibliometric and systematic literature review from 2019-2024. *International Education Trend Issues*, 2(2), 231-255. <https://doi.org/10.56442/ieti.v2i2.647>
- Tahiru, F. (2021). AI in education: A systematic literature review. *Journal of Cases on Information Technology (JCIT)*, 23(1), 1-20. <https://doi.org/10.4018/JCIT.2021010101>
- Tan, S. C., Lee, A. V. Y., & Lee, M. (2022). A systematic review of artificial intelligence techniques for collaborative learning over the past two decades. *Computers and Education: Artificial Intelligence*, 3, Article 100097. <https://doi.org/10.1016/j.caeai.2022.100097>
- Tan, B., Jin, H. Y., & Cutumisu, M. (2024). The applications of machine learning in computational thinking assessments: A scoping review. *Computer Science Education*, 34(2), 193-221. <https://doi.org/10.1080/08993408.2023.2245687>
- Tang, K.Y., Chang, C.Y., & Hwang, G.J. (2023). Trends in artificial intelligence-supported e-learning: A systematic review and co-citation network analysis (1998-2019). *Interactive Learning Environments*, 31(4), 2134-2152. <https://doi.org/10.1080/10494820.2021.1875001>
- Tissenbaum, M., & Slotta, J. D. (2019). Developing a smart classroom infrastructure to support real-time student collaboration and inquiry: A 4-year design study. *Instructional Science*, 47(4), 423-462. <https://doi.org/10.1007/s11251-019-09486-1>
- Vargas-Murillo, A. R., Pari-Bedoya, I. N. M. D. L. A., & Guevara-Soto, F. D. J. (2023, June). The ethics of AI assisted learning: A systematic literature review on the impacts of

- ChatGPT usage in education. In *Proceedings of the 2023 8th International Conference on Distance Education and Learning* (pp. 8-13). Association for Computing Machinery.
<https://doi.org/10.1145/3606094.3606101>
- Wang, T., & Cheng, E. C. (2021, July). Towards a tripartite research agenda: A scoping review of artificial intelligence in education research. In *International Conference on Artificial Intelligence in Education Technology* (pp. 3-24). Singapore: Springer Nature Singapore.
https://doi.org/10.1007/978-981-16-7527-0_1
- Wohlin, C. (2014, May). Guidelines for snowballing in systematic literature studies and a replication in software engineering. In *Proceedings of the 18th international conference on evaluation and assessment in software engineering* (pp. 1-10). Association for Computing Machinery.
<https://doi.org/10.1145/2601248.2601268>
- Wohlin, C., Kalinowski, M., Felizardo, K. R., & Mendes, E. (2022). Successful combination of database search and snowballing for identification of primary studies in systematic literature studies. *Information and Software Technology*, 147, Article 106908.
<https://doi.org/10.1016/j.infsof.2022.106908>
- Wu, R., & Yu, Z. (2024). Do AI chatbots improve students learning outcomes? Evidence from a meta-analysis. *British Journal of Educational Technology*, 55(1), 10-33.
<https://doi.org/10.1111/bjet.13334>
- Xu, W., & Ouyang, F. (2022a). A systematic review of AI role in the educational system based on a proposed conceptual framework. *Education and Information Technologies*, 27(3), 4195-4223. <https://doi.org/10.1007/s10639-021-10774-y>
- Xu, W., & Ouyang, F. (2022b). The application of AI technologies in STEM education: A systematic review from 2011 to 2021. *International Journal of STEM Education*, 9(1), 59.
<https://doi.org/10.1186/s40594-022-00377-5>
- Yue, M., Jong, M. S. Y., & Dai, Y. (2022). Pedagogical design of K-12 artificial intelligence education: A systematic review. *Sustainability*, 14(23), 15620. <https://doi.org/10.3390/su142315620>
- Zafari, M., Bazargani, J. S., Sadeghi-Niaraki, A., & Choi, S. M. (2022). Artificial intelligence applications in K-12 education: A systematic literature review. *Ieee Access*, 10, 61905-61921. <https://doi.org/10.1109/ACCESS.2022.3179356>
- Zhai, X., Yin, Y., Pellegrino, J. W., Haudek, K. C., & Shi, L. (2020). Applying machine learning in science assessment: A systematic review. *Studies in Science Education*, 56(1), 111-151.
<https://doi.org/10.1080/03057267.2020.1735757>
- Zhai, X. (2019, June). Parameters for machine learning. Internal Document: [unpublished].
<https://doi.org/10.13140/RG.2.2.33563.31525>

- Zhang, K., & Aslan, A. B. (2021). AI technologies for education: Recent research & future directions. *Computers and Education: Artificial Intelligence*, 2, Article 100025. <https://doi.org/10.1016/j.caeai.2021.100025>
- Zhang, R., Zou, D., & Cheng, G. (2024). A review of chatbot-assisted learning: Pedagogical approaches, implementations, factors leading to effectiveness, theories, and future directions. *Interactive Learning Environments*, 32(8), 4529-4557. <https://doi.org/10.1080/10494820.2023.2202704>
- Zhou, T., Wu, X., Wang, Y., Wang, Y., & Zhang, S. (2024). Application of artificial intelligence in physical education: A systematic review. *Education and Information Technologies*, 29(7), 8203-8220. <https://doi.org/10.1007/s10639-023-12128-2>

Appendix A. The systematic reviews included in this umbrella review.

Author/Year	Title	Number of Studies Included	Year Range	Country/Region of the First Author	Whether Empirical	Focus
AbdulMuhsen AlHarbi, 2022	The uses of machine learning (ML) in teaching and learning English language: A methodical review.	Unspecified	Unspecified	Saudi Arabia	Unspecified	AI Applications AI Technology Development
Alalawi et al., 2023	Contextualizing the current state of research on the use of machine learning for student performance prediction: A systematic literature review	162	2010-2022	Australia; Saudi Arabia	Unspecified	AI Applications AI Technology Development
Alenezi & Faisal, 2020	Utilizing crowdsourcing and machine learning in education: Literature review.	30	2013- 2019	Kuwait	Unspecified	AI Applications
Almelhes, 2023	A review of artificial intelligence adoption in second-language learning.	39	2011- unspecified	Saudi Arabia	Empirical	AI Applications
Almusaed et al., 2023	Enhancing student engagement: Harnessing “AIED”’s power in hybrid education—A review analysis.	Unspecified	2018-2023	Sweden	Unspecified	AI Applications
Ashok et al., 2021	A systematic survey of cognitive chatbots in personalized learning framework.	40	2017-2021	India	Unspecified	AI Applications
Baranwal, 2022	A systematic review of exploring the potential of teachable agents in English learning	9	2015-2019	Taiwan	Empirical	AI Applications
Bennani & Ben Ghezala, 2022	Adaptive gamification in E-learning: A literature review and future challenges.	Unspecified	2011-2021	Tunisia	Unspecified	AI Applications
Bhutoria, 2022	Personalized education and artificial intelligence in the United States,	353	2019-2021	India, UK, Saudi Arabia	Empirical	AI Applications AI Challenges

	China, and India: A systematic review using a human-in-the-loop model.					
bin Mohamed et al., 2022	Artificial intelligence in mathematics education: A systematic literature review	20	2017-2021	Malaysia	Unspecified	AI Applications
Bozkurt et al., 2021	Artificial intelligence and reflections from educational landscape: A review of AI Studies in half a century.	276	1976-2019	Turkey, South Africa	Unspecified	AI Applications
Casal-Otero et al., 2023	AI literacy in K-12: a systematic literature review	179	2004-2023	Spain	Unspecified	AI Education and Literacy
Celik et al., 2022	The promises and challenges of artificial intelligence for teachers: A systematic review of research.	44	2004-2020	Finland	Empirical	AI Applications AI Challenges AI Technology Development
Chen et al., 2020	Artificial intelligence in education: A review.	30	2019-2010	China	Empirical	AI Applications
Chen et al., 2020	Application and theory gaps during the rise of Artificial Intelligence in Education	45	1999-2019	Hong Kong	Unspecified	AI Technology Development AIED Theories
Chen et al., 2021	Artificial intelligence-assisted personalized language learning: systematic review and co-citation analysis	17	Unspecified	Hong Kong	Empirical	AI Technology Development
Chichekian & Benteux, 2022	The potential of learning with (and not from) artificial intelligence in education	48	2009-2019	Canada	Unspecified	AI Applications
Chiu et al., 2023	Systematic literature review on opportunities, challenges, and future research recommendations of artificial intelligence in education	92	2012-2021	Hong Kong	Empirical	AI Applications AI Challenges
Chng et al., 2023	Examining the use of emerging technologies in schools: A review of artificial intelligence and immersive technologies in STEM education.	82	Unspecified	USA	Empirical	AI Applications

Chu et al., 2022	Artificial intelligence-based robots in education: A systematic review of selected SSCI publications	13	2008-2021	Taiwan	Empirical	AI Applications
Crompton et al., 2024	Affordances and challenges of artificial intelligence in K-12 education: A systematic review	169	2011-2021	USA	Empirical	AI Applications
Cuéllar-Rojas et al., 2022	Bibliometric analysis and systematic literature review of the intelligent tutoring systems	163	1979-2021	Spain, Colombia, Mexico	Unspecified	AI Applications AI Challenges
Dai & Ke, 2022	Educational applications of artificial intelligence in simulation-based learning: A systematic mapping review	59	1998-2021	USA	Mixed	AI Applications AI Technology Development AIED Theories
Deng & Yu, 2022	A systematic review of machine-translation-assisted language learning for sustainable education.	26	2012-2022	China	Unspecified	AI Applications AIED Theories
Deng & Yu, 2023	A meta-analysis and systematic review of the effect of chatbot technology use in sustainable education.	32	2010-2022	China	Empirical	AI Applications
e Silva et al., 2023	Applications of convolutional neural networks in education: A systematic literature review	133	Unspecified	Brazil	Empirical	AI Technology Development
Fang et al., 2023	A systematic review of artificial intelligence technologies used for story writing	27	2018-2022	Hong Kong	Mixed	AI Applications
Fleckenstein et al., 2023	Automated feedback and writing: A multi-level meta-analysis of effects on students' performance	20	Unspecified	Germany	Empirical	AI Applications
García-Martínez et al., 2023	Analysing the impact of artificial intelligence and computational sciences on student performance: Systematic review and meta-analysis.	25	2010-2020	Spain	Empirical	AI Applications

Gennari et al., 2023	How to playfully teach AI to young learners: a systematic literature review.	24	2013-2023	Italy	Unspecified	AI Education and Literacy
Gökçearslan et al., 2024	Benefits, challenges, and methods of artificial intelligence (AI) chatbots in education: A systematic literature review.	37	2020-2023	Turkey	Mixed	AI Applications AI Challenges
González-Calatayud et al., 2021	Artificial intelligence for student assessment: A systematic review.	22	2010-2020	Spain	Empirical	AI Applications AIED Theories
Hahn et al., 2021	A systematic review of the effects of automatic scoring and automatic feedback in educational settings.	125	2016-2020	Spain	Unspecified	AI Applications
Hopcan et al., 2023	Artificial intelligence in special education: A systematic review	29	2008-2020	Turkey	Empirical	AI Applications AI Technology Development
Huang et al., 2023	Trends, research issues and applications of artificial intelligence in language education.	516	2000-2019	Hong Kong	Empirical	AI Applications AI Challenges
Hwang & Chang, 2023	A review of opportunities and challenges of chatbots in education	29	1999-2020	Taiwan	Mixed	AI Applications
Hwang & Tu, 2021	Roles and Research Trends of Artificial Intelligence in Mathematics Education: A Bibliometric Mapping Analysis and Systematic Review	43	Unspecified	Taiwan	Empirical	AI Applications AI Technology Development
Hwang et al., 2022	AI in online-learning research: Visualizing and interpreting the journal publications from 1997 to 2019.	224	1997-2019	Taiwan	Empirical	AI Applications AI Technology Development
Ilić et al., 2023	Intelligent techniques in e-learning: a literature review	305	2010-2021	Serbia	Unspecified	AI Applications AI Challenges
Iyamuremye et al., 2024	Utilization of artificial intelligence and machine learning in chemistry education: a critical review	62	2018-2024	Rwanda	Mixed	AI Applications AI Challenges

Jeon et al., 2024	A systematic review of research on speech-recognition chatbots for language learning: Implications for future directions in the era of large language models	32	2010-2022	USA	Empirical	AI Applications AI Technology Development
Ji et al., 2023	A systematic review of conversational AI in language education: Focusing on the collaboration with human teachers.	24	2015-2021	USA	Empirical	AI Applications AI Challenges
Karan & Angadi, 2023	Artificial intelligence integration into school education: A review of Indian and foreign perspectives.	28	2018-2022	India	Unspecified	AI Education and Literacy AI Applications
Katsarou et al., 2023	A Systematic Review of Voice-based Intelligent Virtual Agents in EFL Education.	10	2015-2020	Greece	Empirical	AI Applications
Klimova et al., 2023	Neural machine translation in foreign language teaching and learning: a systematic review.	13	2016- 2022	Czech Republic	Empirical	AI Applications
Kuhail et al., 2023	Interacting with educational chatbots: A systematic review	36	2011-2021	United Arab Emirates	Empirical	AI Applications AI Challenges AI Technology Development AIED Theories
Liang et al., 2023	Roles and research foci of artificial intelligence in language education: an integrated bibliographic analysis and systematic review approach	55	1990-2020	Taiwan	Empirical	AI Applications AI Technology Development
Lin & Yu, 2024	A bibliometric analysis of artificial intelligence chatbots in educational contexts	384	2006-2023	China	Unspecified	AI Applications
Lin et al., 2023	Artificial intelligence in intelligent tutoring systems toward sustainable education: a systematic review	29	2014-2023	Taiwan	Unspecified	AI Applications AI Challenges AI Technology Development

Lo, 2023	What is the impact of ChatGPT on education? A rapid review of the literature.	50	2022-2023	Hong Kong	Unspecified	AI Applications AI Challenges
Luan & Tsai, 2021	A review of using machine learning approaches for precision education.	40	2016-2020	Taiwan	Empirical	AI Applications AI Technology Development
Luzano, 2024	Assessment in mathematics education in the sphere of artificial intelligence: A systematic review on its threats and opportunities.	38	Unspecified	Philippines	Unspecified	AI Applications AI Challenges
Maita et al., 2024	Pros and cons of artificial intelligence-ChatGPT adoption in education settings: A literature review and future research agendas	45	Unspecified	Indonesia	Mixed	AI Applications AI Challenges AIED Theories
Mallik & Gangopadhyay, 2023	Proactive and reactive engagement of artificial intelligence methods for education: a review	195	2003-2022	USA	Mixed	AI Applications
Martínez-Comesaña et al., 2023	Impact of artificial intelligence on assessment methods in primary and secondary education: Systematic literature review	9	2010-2023	Spain	Empirical	AI Applications
Martinez-Roig et al., 2023	Social robotics in music education: A systematic review	4	2015-2022	Spain	Empirical	AI Applications
Montenegro-Rueda et al., 2023	Impact of the implementation of ChatGPT in education: A systematic review.	12	2022-2023	Spain	Mixed	AI Applications AI Challenges
Moore et al., 2023	What would the matrix do?: A systematic review of K-12 AI learning contexts and learner-interface interactions.	12	2014-2021	USA	Empirical	AI Applications
Mousavinasab et al., 2021	Intelligent tutoring systems: a systematic review of characteristics, applications, and evaluation methods	53	2007-2017	Iran	Empirical	AI Applications AI Technology Development

Murtaza et al., 2022	AI-based personalized e-learning systems: Issues, challenges, and solutions.	152	2010-2022	Pakistan	Empirical	AI Applications AI Challenges AI Technology Development
Neumann, 2020	Social robots and young children's early language and literacy learning	13	2004-2018	Australia	Empirical	AI Applications AIED Theories
Ng et al., 2023	A review of AI teaching and learning from 2000 to 2020.	49	2000-2020	China	Empirical	AI Applications
Okonkwo & Ade-Ibijola, 2021	Chatbots applications in education: A systematic review	53	Unspecified -2021	South Africa	Empirical	AI Applications AI Challenges
Orji & Vassileva, 2022	Automatic modeling of student characteristics with interaction and physiological data using machine learning: A review	38	2015-2022	Canada	Empirical	AI Applications AI Technology Development
Oyelere et al., 2022	Artificial intelligence in African schools: Towards a contextualized approach.	54	Unspecified	Sweden	Unspecified	AI Education and Literacy
Ramesh & Sanampudi, 2022	An automated essay scoring systems: a systematic literature review	62	2010-2020	India	Empirical	AI Applications AI Challenges AI Technology Development
Reindl, 2021	Emotion AI in education: a literature review	12	2005-2020	China	Mixed	AI Applications
Rice & Dunn, 2023	The use of artificial intelligence with students with identified disabilities: A systematic review with critique.	18	2009-2022	USA	Empirical	AI Applications
Riza et al., 2023	Use of chatbots in e-learning context: A systematic review.	20	2017-2022	Indonesia	Unspecified	AI Applications
Salas-Pilco et al., 2022	Artificial intelligence and learning analytics in teacher education: A systematic review.	30	2017-2021	China	Empirical	AI Education and Literacy AI Applications AI Technology Development
Salas-Pilco et al., 2022	Artificial Intelligence and New Technologies in Inclusive Education	27	2017-2021	China	Empirical	AI Applications AI Challenges

	for Minority Students: A Systematic Review					
Sanusi et al., 2023	A systematic review of teaching and learning machine learning in K-12 education	43	Unspecified	Finland	Unspecified	AI Education and Literacy AI Challenges
Saputra et al., 2023	Integration of artificial intelligence in education: Opportunities, challenges, threats and obstacles. A literature review.	30	Unspecified	Indonesian	Unspecified	AI Challenges
Sharadgah & Sa'di, 2022	A systematic review of research on the use of artificial intelligence in English language teaching and learning (2015-2021): What are the current effects?.	64	2015-2021	Saudi Arabia	Empirical	AI Applications AI Challenges
Sikström et al., 2022	How pedagogical agents communicate with students: A two-phase systematic review.	37	2010-2020	Finland	Mixed	AI Applications AIED Theories
Su & Yang, 2022	Artificial intelligence in early childhood education: A scoping review	17	1995-2021	China	Unspecified	AI Applications
Su et al., 2022	A meta-review of literature on educational approaches for teaching AI at the K-12 levels in the Asia-Pacific region	14	2018-2021	Hong Kong	Empirical	AI Education and Literacy
Su et al., 2023	Artificial intelligence (AI) literacy in early childhood education: The challenges and opportunities.	16	2016-2022	Hong Kong	Mixed	AI Education and Literacy AI Applications
Su et al., 2024	Teaching artificial intelligence in K-12 classrooms: a scoping review	21	2019-2022	Hong Kong	Empirical	AI Education and Literacy AI Applications
Suryanti et al., 2024	Artificial Intelligence in Education: Bibliometric and Systematic Literature Review from 2019-2024	98	2019-2024	Indonesia	Unspecified	AI Applications AI Challenges

Tahiru, 2021	AI in education: A systematic literature review.	23	2010-2019	Ghana	Empirical	AI Applications AI Challenges
Tan et al., 2022	A systematic review of artificial intelligence techniques for collaborative learning over the past two decades	41	2002-2022	Singapore	Unspecified	AI Applications AI Technology Development
Tan et al., 2024	The applications of machine learning in computational thinking assessments: A scoping review	20	Unspecified	Canada	Empirical	AI Applications AI Technology Development
Tang et al., 2023	Trends in artificial intelligence-supported e-learning: A systematic review and co-citation network analysis (1998-2019).	86	Unspecified	Taiwan	Unspecified	AI Applications
Vargas-Murillo et al., 2023	The ethics of AI assisted learning: A systematic literature review on the impacts of ChatGPT usage in education.	16	2023	Peru	Unspecified	AI Education and Literacy AI Applications AI Challenges
Weng & Chiu, 2023	Instructional design and learning outcomes of intelligent computer assisted language learning: Systematic review in the field	83	Unspecified	Hong Kong	Empirical	AI Applications AI Challenges
Wu & Yu, 2024	Do AI chatbots improve students learning outcomes? Evidence from a meta-analysis	24	Unspecified	China	Empirical	AI Applications
Xu & Margevica-Grinberga, 2021	A discourse on innovation of English teaching in China from the perspective of artificial intelligence	81	2011-2021	Latvia	Empirical	AI Applications
Xu & Ouyang, 2022	A systematic review of AI role in the educational system based on a proposed conceptual framework.	51	2005-2021	China	Mixed	AI Applications

Xu & Ouyang, 2022	The application of AI technologies in STEM education: a systematic review from 2011 to 2021	63	2011-2021	China	Empirical	AI Applications
Yang & Kyun, 2022	The current research trend of artificial intelligence in language learning: A systematic empirical literature review from an activity theory perspective.	25	2007-2021	Australia	Empirical	AI Applications
Younis et al., 2023	A systematic literature review on the applications of robots and natural language processing in education.	82	2014-2023	Malaysia, Iraq	Empirical	AI Applications
Yue et al., 2022	Pedagogical design of K-12 artificial intelligence education: A systematic review.	32	2010-2022	China	Empirical	AI Education and Literacy
Zafari et al., 2022	Artificial intelligence applications in K-12 education: A systematic literature review.	210	2011-2021	Iran, Korea	Empirical	AI Applications AI Technology Development
Zhai & Wibowo, 2022	A systematic review on cross-culture, humor and empathy dimensions in conversational chatbots: the case of second language acquisition.	48	2012-2022	Australia	Empirical	AI Applications
Zhai et al., 2020	Applying machine learning in science assessment: a systematic review	49	2008-2018	USA	Empirical	AI Applications AI Technology Development
Zhai et al., 2021	A Review of Artificial Intelligence (AI) in Education from 2010 to 2020.	100	2010-2020	China	Mixed	AI Applications
Zhan et al., 2024	A systematic literature review of game-based learning in Artificial Intelligence education	125	1999-2021	China	Empirical	AI Education and Literacy AI Applications
Zhang & Aslan, 2021	AI technologies for education: Recent research & future directions	40	1999-2020	USA	Empirical	AI Applications
Zhang et al., 2024	A review of chatbot-assisted learning: pedagogical approaches, implementations, factors leading to	46	2000-2022	Hong Kong	Empirical	AI Applications

	effectiveness, theories, and future directions					
Zheng et al., 2023	The effectiveness of artificial intelligence on learning achievement and learning perception: A meta-analysis	24	2001-2020	China	Empirical	AI Applications
Zhou et al., 2024	Application of artificial intelligence in physical education: a systematic review	130	2010-2023	Republic of Korea	Empirical	AI Applications AI Challenges AI Technology Development